POLAR: Fractional Spectral Link Recommendation via Pole-Shared Krylov Sketching

(Extended Version with Complete Proofs)

***Abstract*—Spectral link recommendation improves a graph by selecting a small set of new edges. We study the regularized fractional inverse trace $\mathrm{Tr}((L+\sigma I)^{-s})$, $0<s<1$, whose gain averages heat-trace reductions over diffusion times of mean s/σ, interpolating between degree heuristics and effective resistance. Optimizing this objective requires efficient marginal evaluation despite nonlinear matrix powers, non-submodularity, and repeated linear solves. We present POLAR (Pole-shared Orthogonalized Lazy Algebraic Re-ranking). A positive Stieltjes quadrature yields a multi-pole Sherman–Morrison formula with dimension-free relative marginal error and $O((1+\log\kappa_*)\log(1/\epsilon))$ poles. Explicit submodularity-ratio and curvature bounds establish an error-aware greedy guarantee. The algorithm shares Krylov iterations across poles, maintains endpoint sketches across rounds, and re-ranks a proxy shortlist algebraically, so k rounds need $2q+kB$ rather than $2kq$ multi-shift solves. On eighteen real networks with up to 4.0 million vertices and 34.7 million edges, POLAR attains the exhaustive optimum on all small instances and 99.2–100% of exact fractional greedy on five medium networks, while stochastic Lanczos greedy attains 93.3–99.5% at 12–$39\times$ the time. With $B=1$, it is 8.9–$17.8\times$ faster than sketch recomputation with identical selections and processes LiveJournal in 15 minutes. On a multiscale benchmark with $\sigma=10^{-2}$, its gains exceed exact inverse-trace greedy by 48.06% and 21.77% at $s=0.25$ and $s=0.5$.**

***Index Terms*—Spectral graph optimization, link recommendation, fractional matrix functions, rational quadrature, Krylov methods, non-submodular optimization.**

I. Introduction

Graph augmentation recommends new links to improve a structural objective under a limited budget. Effective-resistance minimization supports continuous edge-weight design [1] and discrete augmentation [2]; its regularized analogue minimizes $\mathrm{Tr}((L+\sigma I)^{-1})$. Related graph-mining tasks include current-flow group selection [3], [4] and forest-matrix queries on evolving graphs [5]. Fractional graph operators provide a continuous choice of spectral weighting [6].

We optimize the fractional gain

$$f_s(S)=\mathrm{Tr}(M_0^{-s})-\mathrm{Tr}((M_0+L_S)^{-s}),\quad 0<s<1, \quad (1)$$

where $M_0=L_0+\sigma I$ and L_S contains the selected unit-weight links. The exponent selects a diffusion time scale. By Proposition 1, $f_s(S)$ averages the reduction of the heat trace $\mathrm{Tr}\,e^{-tL}$, the total return probability of continuous-time random walks, over times t with a Gamma weight of mean s/σ. Degree heuristics maximize this reduction as $t\to 0$, whereas inverse-trace (effective-resistance) augmentation, used for example to rewire graph neural networks against over-squashing [7], is the case $s=1$, the longest horizon in this family. The pair (s,σ) therefore lets an analyst target the horizon at which spreading, consensus, or message passing is observed. On real networks with $\sigma=1$, exact ER, log-det, and fractional greedy reach values of f_s within a few percent, yet the $s=0.25$ and ER selections have Jaccard similarity only 0.43–0.74 on three medium networks, and on ca-HepPh the log-det and ER selections differ by 24% in heat-trace reduction at $t=3.2$. On a multiscale graph where short and long horizons conflict, fractional selection exceeds ER greedy by up to 48% (Fig. 1(b)).

Three obstacles arise when turning this objective into an efficient recommendation algorithm. *First, fractional matrix powers lack the inverse's direct rank-one correction.* Approximating two traces separately to scalar accuracy η gives only a $2n\eta$ absolute marginal-error bound. We instead need accuracy relative to the marginal itself. *Second, the gain is not generally submodular.* A four-vertex example exhibits increasing marginal returns, invalidating the stale upper bounds used by classical CELF evaluation [8]. *Third, linear algebra repeats across poles and rounds.* Recomputing two sketch banks of dimension q in each of k rounds costs $2kq$ multi-shift solves and $O(kpqN)$ candidate scoring with p poles and N candidates. Shift invariance shares the matrix products across poles [9]; exploiting the evolving graph requires additional structure.

We present POLAR, *Pole-shared Orthogonalized Lazy Algebraic Re-ranking*, with four contributions.

1) **Accurate marginal algebra (first obstacle).** Positive Stieltjes quadrature converts a fractional marginal into a sum of rank-one inverse updates. Relative derivative accuracy, integrated along the update path, gives an n-independent marginal guarantee with $O((1+\log\kappa_*)\log(1/\epsilon))$ poles.
2) **Guarantees beyond submodularity (second obstacle).** Explicit, σ-dependent submodularity-ratio and curvature bounds instantiate the ratio–curvature framework [10] with quadrature, sketch, and solve errors, and valid upper envelopes restore lazy evaluation without diminishing returns. The worst-case factor is conservative, so we also report the measured per-round factor.
3) **Reuse across poles, rounds, and candidates (third obstacle).** Unlike the positive-power trace of FastFrac [11], negative exponents require squared resolvent norms, which we sketch. One selected-edge solve updates all poles and endpoint sketches, norm caches cost

$O(pn_{\mathcal{C}}q+pN)$ per round, and algebraic re-ranking reuses the champion's solve, so k rounds need $2q+kB$ instead of $2kq$ multi-shift solves.

4) **Evaluation.** On eighteen networks with up to 34.7 million edges, POLAR matches exhaustive search on small instances, attains 99.2–100% of exact fractional greedy ($B = 16$), and is 8.9–17.8× faster than sketch recomputation with identical selections ($B = 1$). Heat-trace curves confirm that s orders the selected links by diffusion horizon.

II. Preliminaries, Problem Statement, and Spectral Motivation

A. Graph Model and Optimization Objectives

Let $G_0 = (V, E_0, w_0)$ be an undirected graph with n vertices and nonnegative edge weights. Write L_0 for its Laplacian and $d_{\max,0}$ for its maximum weighted degree. Connectivity is not required. Let $\mathcal{C}$ contain N distinct candidate non-loop edges not in E_0, and let $1 \le k \le N$. Every selected candidate is added with unit weight. For $e = \{u, v\}$, choose either orientation and define $b_e = \boldsymbol{e}_u - \boldsymbol{e}_v$, so $\|b_e\|_2^2 = 2$.

The active endpoints are $V_{\mathcal{C}} = \bigcup_{e\in\mathcal{C}} \operatorname{supp}(b_e)$, with $n_{\mathcal{C}} = |V_{\mathcal{C}}| \le \min\{n, 2N\}$. We write p for the number of quadrature poles, q for the dimension of each sketch bank, B for the re-ranked shortlist size, and t for the maximum number of CG iterations per right-hand side.

For $S \subseteq \mathcal{C}$, define

$$\begin{aligned} L_S &= \sum_{e\in S} b_e b_e^\top, \\ M_S &= M(S) = L_0 + \sigma I + L_S, \qquad \sigma > 0. \end{aligned} \tag{2}$$

We use $M_0 = M_\varnothing$. Fractional powers and logarithms are defined by spectral calculus for positive definite matrices. The two normalized objectives are

$$f_s(S) = \operatorname{Tr}(M_0^{-s}) - \operatorname{Tr}(M_S^{-s}), \quad 0 < s < 1, \tag{3}$$

$$\begin{aligned} f_{\log}(S) &= \operatorname{Tr}(\log M_S) - \operatorname{Tr}(\log M_0) \\ &= \log\det M_S - \log\det M_0. \end{aligned} \tag{4}$$

The optimization problem is

$$\max_{S\subseteq\mathcal{C},\ |S|\le k} f_s(S). \tag{5}$$

The log-determinant gain provides the comparison objective in Proposition 2. At $s = 1$ and $\sigma = 1$, the inverse is the forest matrix, which supports forest closeness and dynamic proximity queries [5], [12]. Laplacian resolvents also appear in Friedkin–Johnsen opinion quantities [13]. The choice of spectral function matters: Kemeny's constant can increase after an edge insertion [14], whereas the inverse-trace objectives considered here strictly decrease.

The positive marginal and the negative trace change are denoted by

$$g_s(e \mid S) = f_s(S \cup \{e\}) - f_s(S), \tag{6}$$

$$\begin{aligned} \Delta_s(e; M_S) &= \operatorname{Tr}((M_S + b_e b_e^\top)^{-s}) - \operatorname{Tr}(M_S^{-s}) \\ &= -g_s(e \mid S). \end{aligned} \tag{7}$$

Let O denote an optimal budget-k set.

All comparison sets used below contain at most $2k$ added edges. For such sets,

$$\sigma I \preceq M_S \preceq \Lambda I, \qquad \Lambda = \sigma + 2d_{\max,0} + 4k, \qquad \kappa_* = \Lambda/\sigma. \tag{8}$$

Indeed, for every $x \in \mathbb{R}^n$,

$$\begin{aligned} x^\top L_0 x &= \sum_{\{u,v\}\in E_0} w_{uv}(x_u - x_v)^2 \\ &\le 2\sum_v d_v x_v^2 \le 2d_{\max,0}\|x\|_2^2, \end{aligned}$$

and

$$\|L_S\|_2 \le \sum_{e\in S} \|b_e b_e^\top\|_2 = 2|S| \le 4k.$$

The same interval contains an edge-addition path whose endpoint has at most $2k$ selected edges.

B. Spectral Sensitivity and Budget Allocation

For a candidate e, consider $M_t = M_S + t b_e b_e^\top$, $0 \le t \le 1$. Let $\lambda_i(t)$ and $v_i(t)$ be an orthonormal eigensystem of $L_0 + L_S + t b_e b_e^\top$.

Lemma 1 (Spectral sensitivity)**.** *For every $0 < s < 1$,*

$$\begin{aligned} g_s(e \mid S) &= s\int_0^1 b_e^\top M_t^{-s-1} b_e \, dt \\ &= \int_0^1 \sum_{i=1}^n \frac{s(v_i(t)^\top b_e)^2}{(\lambda_i(t)+\sigma)^{s+1}}\, dt. \end{aligned} \tag{9}$$

In particular, every candidate has a strictly positive marginal.

Proof. Set $c_s = \sin(\pi s)/\pi > 0$. The Stieltjes identity, derived explicitly in (14), gives

$$\operatorname{Tr}(M_t^{-s}) = c_s \int_0^\infty \tau^{-s} \operatorname{Tr}((M_t + \tau I)^{-1})\, d\tau,$$

Differentiating an inverse and using cyclic invariance of trace gives

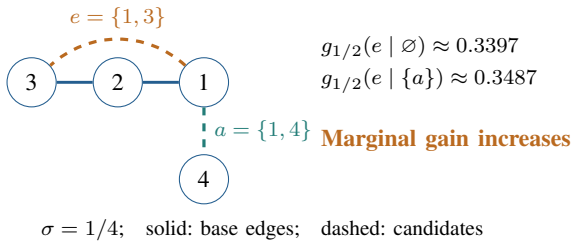
$$\frac{d}{dt}\operatorname{Tr}((M_t+\tau I)^{-1}) = -b_e^\top (M_t + \tau I)^{-2} b_e.$$

The absolute value is at most $2/(\sigma+\tau)^2$. Its product with τ^{-s} is integrable at zero because $s < 1$ and at infinity because $s > 0$. Differentiation under the integral is therefore valid. Differentiating the scalar Stieltjes identity and applying it to each eigenvalue of M_t yields

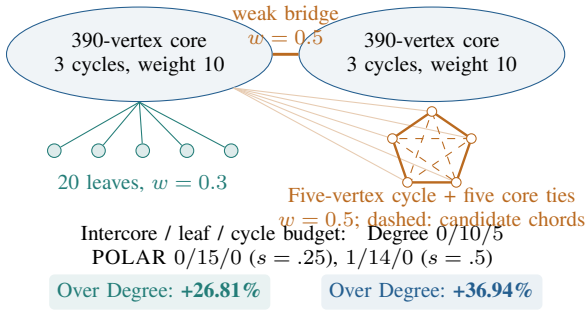
$$\begin{aligned} -\frac{d}{dt}\operatorname{Tr}(M_t^{-s}) &= c_s \int_0^\infty \tau^{-s} b_e^\top (M_t+\tau I)^{-2} b_e \, d\tau \\ &= s b_e^\top M_t^{-s-1} b_e. \end{aligned}$$

Integrating from zero to one proves the first equality in (9). Diagonalizing M_t proves the second. The quadratic form is positive because $M_t^{-s-1} \succ 0$ and $b_e \neq 0$, proving strict positivity. □

For a simple eigenvalue, differentiating its normalized eigenvector equation also gives $\dot\lambda_i(t) = (v_i(t)^\top b_e)^2$: multiply



(a) An exact four-vertex counterexample to diminishing returns (Example 1).



(b) Heterogeneous multiscale graph, $n = 805$. Improvements are three-seed mean gains at $s = .25$ (left) and $s = .5$ (right).

Fig. 1. Why fractional spectral optimization needs both a distinct objective and non-submodular evaluation. (a) A candidate becomes more valuable after another edge is inserted. (b) Strongly coupled cores, a weak bridge, peripheral leaves, and an independently anchored cycle separate degree and spectral rankings. POLAR uses $q = 24$, $B = 4$; its gains also exceed Exact-ER by 48.06% and 21.77%, respectively.

the differentiated equation by $v_i(t)^\top$, cancel the eigenvector-derivative terms, and use $v_i(t)^\top v_i(t) = 1$. Thus the scalar sensitivity is

$$w_s(\lambda) = s(\lambda + \sigma)^{-s-1}. \tag{10}$$

Equation (9) remains valid at repeated eigenvalues, where a particular eigenvector derivative need not be defined.

Compared with $s = 1$, a smaller exponent flattens this weight and attenuates the relative emphasis on the lowest nonconstant modes.

Diffusion time scales: The exponent has a direct diffusion interpretation. Let $\Delta H_S(t) = \operatorname{Tr} e^{-tL_0} - \operatorname{Tr} e^{-t(L_0+L_S)} \geq 0$ be the reduction of the heat trace, i.e., of the total return probability of continuous-time random walks after time t.

Proposition 1 (Heat-trace representation)**.** *For $0 < s < 1$,*

$$f_s(S) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-\sigma t} \Delta H_S(t)\, dt, \tag{11}$$
$$f_{\log}(S) = \int_0^\infty t^{-1} e^{-\sigma t} \Delta H_S(t)\, dt,$$

and the same formula with $s = 1$ gives the inverse-trace gain. As $t \to 0$, $\Delta H_S(t) = 2|S|t - \frac{t^2}{2}\Delta_S \operatorname{Tr}(L^2) + O(t^3)$, where adding $\{u, v\}$ increases $\operatorname{Tr}(L^2)$ by $2(d_u + d_v) + 4$.

Proof. Apply $\Gamma(s)x^{-s} = \int_0^\infty t^{s-1}e^{-tx}dt$ and $\log(y/x) = \int_0^\infty t^{-1}(e^{-tx} - e^{-ty})\, dt$ to the eigenvalues of M_0 and M_S, and use $\operatorname{Tr} e^{-tM} = e^{-\sigma t} \operatorname{Tr} e^{-tL}$. Weyl's inequality gives $\Delta H_S \geq 0$. The expansion follows from $\operatorname{Tr} e^{-tL} = n - t \operatorname{Tr} L + \frac{t^2}{2} \operatorname{Tr} L^2 + O(t^3)$, $\operatorname{Tr} L = \sum_v d_v$, and $\operatorname{Tr} L^2 = \sum_v d_v^2 + \sum_{u \neq v} w_{uv}^2$. □

The weight $t^{s-1}e^{-\sigma t}$ is proportional to a Gamma density with mean s/σ. Degree, which minimizes the current endpoint-degree sum, is the greedy maximizer of $\Delta H_S(t)$ as $t \to 0$; log-determinant is the limit of f_s/s as $s \to 0$; and the inverse trace is $s = 1$. The pair (s, σ) is thus a single horizon parameter: decreasing s or increasing σ moves the emphasis to short diffusion times, and Degree, log-determinant, f_s with increasing s, and the inverse trace target increasingly long horizons. Section V-E verifies this order on real networks.

C. An Exact Obstruction to Submodularity

A set function is submodular if its marginal satisfies $g(e \mid A) \geq g(e \mid B)$ whenever $A \subseteq B$ and $e \notin B$. The following instance disproves this property for the fractional objective in general.

Example 1 (A four-vertex obstruction to submodularity)**.** Let $V = \{1, 2, 3, 4\}$, $E_0 = \{\{1, 2\}, \{2, 3\}\}$, $\sigma = 1/4$, and $s = 1/2$. All base and added edges have unit weight. Let $e = \{1, 3\}$ and $a = \{1, 4\}$. For the normalized gain f_s, its marginals obey

$$g_s(e \mid \varnothing) = \frac{2}{\sqrt{5}} - \frac{2}{\sqrt{13}}, \tag{12}$$
$$g_s(e \mid \{a\}) = \frac{8\sqrt{2}}{7} + \frac{2}{3} - \frac{2}{\sqrt{5}} - \frac{2}{\sqrt{13}} - \frac{2}{\sqrt{17}}, \tag{13}$$

and $g_s(e \mid \{a\}) - g_s(e \mid \varnothing) > 1/375 > 0$. Thus f_s is not submodular, even for unweighted base graphs and unit-weight additions.

Proof. The base graph is a three-vertex path and an isolated vertex, so its Laplacian has eigenvalues $0, 0, 1, 3$. Indeed, $(1, 1, 1, 0)^\top$ and $(0, 0, 0, 1)^\top$ are zero eigenvectors, while $(1, 0, -1, 0)^\top$ and $(1, -2, 1, 0)^\top$ have eigenvalues 1 and 3. Adding e gives a triangle and an isolated vertex, with eigenvalues $0, 0, 3, 3$: every vector supported on the first three coordinates and summing to zero has eigenvalue 3.

Adding a to the base graph gives the path 3–2–1–4, whose Laplacian is

$$L_a = \begin{pmatrix} 2 & -1 & 0 & -1 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix}.$$

The subspaces $\{(x, x, y, y)^\top\}$ and $\{(x, -x, y, -y)^\top\}$ are invariant. On these subspaces, the coordinate matrices are respectively $\left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 3 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. Their eigenvalues are $0, 2$ and $2 - \sqrt{2}, 2 + \sqrt{2}$, respectively. After adding both edges, the graph is a triangle with a pendant vertex. Its eigenvectors $(1, 1, 1, 1)^\top$, $(0, 1, 1, -2)^\top$, $(0, 1, -1, 0)^\top$, and $(-3, 1, 1, 1)^\top$

have eigenvalues $0, 1, 3, 4$. These four independent vectors give the complete spectrum.

Adding σI shifts every eigenvalue by $1/4$. Subtracting the trace of the inverse square root after adding e from the trace before adding it therefore gives $g_s(e \mid \varnothing) = 2/\sqrt{5} - 2/\sqrt{13}$. For the second marginal, the three nonzero path eigenvalues contribute

$$\frac{2}{\sqrt{9-4\sqrt{2}}} + \frac{2}{3} + \frac{2}{\sqrt{9+4\sqrt{2}}}.$$

Both terms in the following sum are positive, and

$$\left(\frac{1}{\sqrt{9-4\sqrt{2}}} + \frac{1}{\sqrt{9+4\sqrt{2}}}\right)^2 = \frac{18}{49} + \frac{2}{7} = \frac{32}{49}.$$

Consequently the sum equals $4\sqrt{2}/7$. The common zero-eigenvalue contribution cancels, yielding the stated second marginal. Their difference is

$$D = \frac{8\sqrt{2}}{7} + \frac{2}{3} - \frac{4}{\sqrt{5}} - \frac{2}{\sqrt{17}}.$$

The inequalities $707^2 < 2 \cdot 500^2$, $16 \cdot 10000 < 5 \cdot 179^2$, and $4 \cdot 10000 < 17 \cdot 49^2$ imply

$$\sqrt{2} > \frac{707}{500}, \qquad \frac{4}{\sqrt{5}} < \frac{179}{100}, \qquad \frac{2}{\sqrt{17}} < \frac{49}{100}.$$

Substitution gives

$$D > \frac{8}{7}\frac{707}{500} + \frac{2}{3} - \frac{179}{100} - \frac{49}{100} = \frac{1}{375} > 0.$$

This strictly reverses the diminishing-returns inequality for $\varnothing \subset \{a\}$ and $e \notin \{a\}$. □

Numerically, the two marginals are 0.3397269948 and 0.3487121007; the strict inequality above is certified without floating-point arithmetic.

III. Quadrature-Based Marginal Formula and Approximation Guarantees

This extended version of the conference paper contains the complete proofs of all results.

A. Positive Composite Quadrature

Contour-integral quadrature and conformal maps provide rapidly convergent rational approximations to matrix powers and logarithms [15]. We use a positive Stieltjes representation so that every pole contributes a nonnegative marginal. The approximation below controls a derivative along the update path and therefore avoids subtracting two approximate traces.

Let $c_s = \sin(\pi s)/\pi$ for $0 < s < 1$. For $x > 0$, the Stieltjes representation is

$$x^{-s} = c_s \int_0^\infty \frac{\tau^{-s}}{x+\tau}\, d\tau. \tag{14}$$

Indeed, substituting $\tau = xu$ reduces the right-hand side to

$$c_s x^{-s} \int_0^\infty \frac{u^{-s}}{1+u}\, du = c_s x^{-s} \Gamma(1-s)\Gamma(s) = x^{-s},$$

where the last equality is Euler's reflection formula.

Fix $0 < a \le b$, and define

$$K = \lceil \log_2(b/a) \rceil, \qquad C = 2^K a, \qquad L_\ell = 2^\ell a.$$

When $K = 0$, there are no middle panels. We partition the integral in (14) into $[0, a]$, the intervals $[L_\ell, 2L_\ell]$ for $0 \le \ell < K$, and $[C, \infty)$. For an integer $r \ge 1$, let

$$\{(\ell_i, A_i)\}_{i=1}^r, \quad \{(\xi_i, W_i)\}_{i=1}^r, \quad \{(v_i, B_i)\}_{i=1}^r$$

be the r-point Gaussian rules for the measures $u^{-s}\, du$ on $[0,1]$, du on $[-1,1]$, and $u^{s-1}\, du$ on $[0,1]$, respectively. All weights are strictly positive, all nodes are interior, and each rule integrates polynomials of degree at most $2r-1$ exactly [16].

These rules can be specified explicitly by Jacobi polynomials [17]. Write $P_m^{(\alpha,\beta)}$ for the standard Jacobi polynomial normalized by $P_m^{(\alpha,\beta)}(1) = \binom{m+\alpha}{m}$. Let $z_i^{(\alpha,\beta)}$ be the ith root of $P_r^{(\alpha,\beta)}$, and define

$$h_m^{(\alpha,\beta)} = \frac{2^{\alpha+\beta+1}}{2m+\alpha+\beta+1} \frac{\Gamma(m+\alpha+1)\Gamma(m+\beta+1)}{m!\,\Gamma(m+\alpha+\beta+1)}.$$

The Christoffel weights are

$$w_i^{(\alpha,\beta)} = \left[\sum_{m=0}^{r-1} \frac{P_m^{(\alpha,\beta)}(z_i^{(\alpha,\beta)})^2}{h_m^{(\alpha,\beta)}}\right]^{-1}.$$

The required nodes and weights are

$$\begin{aligned} \ell_i &= \frac{1+z_i^{(0,-s)}}{2}, & A_i &= 2^{s-1} w_i^{(0,-s)}, \\ \xi_i &= z_i^{(0,0)}, & W_i &= w_i^{(0,0)}, \\ v_i &= \frac{1+z_i^{(0,s-1)}}{2}, & B_i &= 2^{-s} w_i^{(0,s-1)}. \end{aligned}$$

All Jacobi parameters used here exceed -1.

The low-tail poles and weights are

$$\tau_i^{\mathrm{lo}} = a\ell_i, \qquad \omega_i^{\mathrm{lo}} = c_s a^{1-s} A_i. \tag{15}$$

For the middle panel $[L_\ell, 2L_\ell]$, they are

$$\begin{aligned} \tau_{\ell i}^{\mathrm{mid}} &= \frac{L_\ell}{2}(3+\xi_i), \\ \omega_{\ell i}^{\mathrm{mid}} &= c_s \frac{L_\ell}{2} W_i \big(\tau_{\ell i}^{\mathrm{mid}}\big)^{-s}. \end{aligned} \tag{16}$$

For the high tail, substituting $\tau = C/u$ gives

$$\int_C^\infty \frac{\tau^{-s}}{x+\tau}\, d\tau = C^{1-s} \int_0^1 \frac{u^{s-1}}{C+xu}\, du.$$

Its poles and weights are consequently

$$\tau_i^{\mathrm{hi}} = \frac{C}{v_i}, \qquad \omega_i^{\mathrm{hi}} = c_s C^{1-s} \frac{B_i}{v_i}. \tag{17}$$

Enumerating these three collections produces $p = (K+2)r$ strictly positive poles and weights and the rational function

$$r_p(x) = \sum_{j=1}^p \frac{\omega_j}{x+\tau_j}. \tag{18}$$

The two tails are transformed, rather than truncated.

B. Dimension-Free Relative Marginal Error

Lemma 2 (A positive Gaussian quadrature estimate)**.** *Let μ be a finite positive measure on $[-1,1]$ and let Q_r be a positive r-point Gaussian rule for μ. Suppose that F is analytic in a neighborhood of $\{z : |z| \le 2\}$, that $|F(z)| \le M$ on $|z| = 2$, and that $F(u) \ge m > 0$ on $[-1,1]$. Then*

$$\left| \int F\, d\mu - Q_r[F] \right| \le \frac{4M}{m} 4^{-r} \int F\, d\mu. \tag{19}$$

Moreover, for $\ell(u) = (1+u)/2$, provided $\int \ell\, d\mu > 0$,

$$\left| \int \ell F\, d\mu - Q_r[\ell F] \right| \le \frac{8M}{m} 4^{-r} \int \ell F\, d\mu. \tag{20}$$

Proof. Write the Taylor series $F(z) = \sum_{\nu=0}^{\infty} a_\nu z^\nu$. Cauchy's coefficient estimate on $|z| = 2$ gives $|a_\nu| \le M 2^{-\nu}$. For $P(u) = \sum_{\nu=0}^{2r-1} a_\nu u^\nu$ and $u \in [-1,1]$,

$$|F(u) - P(u)| \le M \sum_{\nu=2r}^{\infty} 2^{-\nu} = 2M 4^{-r}.$$

Exactness on P and on constants, together with positivity of the weights, yields

$$\begin{aligned} \left| \int F\, d\mu - Q_r[F] \right| &= \left| \int (F-P)\, d\mu - Q_r[F-P] \right| \\ &\le 2M 4^{-r} \big(\mu([-1,1]) + Q_r[1] \big) \\ &= 4M 4^{-r} \mu([-1,1]). \end{aligned}$$

Since $\int F\, d\mu \ge m\mu([-1,1])$, this proves (19).

For the second assertion, let $P_-(u) = \sum_{\nu=0}^{2r-2} a_\nu u^\nu$. Then

$$\sup_{[-1,1]} |F - P_-| \le M \sum_{\nu=2r-1}^{\infty} 2^{-\nu} = 4M 4^{-r}.$$

The polynomial ℓP_- has degree at most $2r-1$. Since $\ell \ge 0$ and $Q_r[\ell] = \int \ell\, d\mu$, we obtain

$$\begin{aligned} \left| \int \ell F\, d\mu - Q_r[\ell F] \right| &\le 4M 4^{-r} \left(\int \ell\, d\mu + Q_r[\ell] \right) \\ &= 8M 4^{-r} \int \ell\, d\mu \\ &\le \frac{8M}{m} 4^{-r} \int \ell F\, d\mu. \end{aligned}$$

This is (20). □

Lemma 3 (Uniform relative derivative approximation)**.** *For (18), define*

$$D(x) = s x^{-s-1}, \qquad D_p(x) = \sum_{j=1}^{p} \frac{\omega_j}{(x+\tau_j)^2}.$$

For every $x \in [a,b]$,

$$|D_p(x) - D(x)| \le 256\, 4^{-r} D(x). \tag{21}$$

Proof. Differentiating (14) under the integral gives

$$D(x) = c_s \int_0^{\infty} \frac{\tau^{-s}}{(x+\tau)^2}\, d\tau. \tag{22}$$

The integrable function $\tau^{-s}/(a+\tau)^2$ justifies this differentiation uniformly for $x \in [a,b]$. We bound the relative error separately on each positive integral piece.

For the low tail, put $\tau = au$ and $v = 2u - 1$. Apart from the positive factor $c_s a^{1-s}(x + a/2)^{-2}$, the kernel against $u^{-s}\, du$ is

$$F_{\mathrm{lo}}(v) = (1+\theta v)^{-2}, \qquad \theta = \frac{a}{2x+a} \le \frac{1}{3}.$$

On $|z| \le 2$, $|1 + \theta z| \ge 1 - 2\theta \ge 1/3$, so the kernel is analytic on a neighborhood of this disk and has boundary modulus at most 9. For real $v \in [-1,1]$,

$$F_{\mathrm{lo}}(v) \ge (1+\theta)^{-2} \ge \frac{9}{16}.$$

Equation (19), applied to the affine image of the measure $u^{-s}\, du$, therefore gives the low-tail relative error $64\, 4^{-r}$.

On a middle panel $[L, 2L]$, substitute $\tau = L(3+v)/2$. The positive prefactor is

$$c_s \frac{L}{2} \left(\frac{3L}{2} \right)^{-s} \left(x + \frac{3L}{2} \right)^{-2},$$

and the remaining kernel is

$$F_{\mathrm{mid}}(v) = (1 + v/3)^{-s} (1 + \theta v)^{-2}, \quad \theta = \frac{L}{2x+3L} \le \frac{1}{3}.$$

The branch positive on the real interval is analytic on $|z| \le 2$, because $\mathrm{Re}(1 + z/3) \ge 1/3$. On its boundary,

$$|F_{\mathrm{mid}}(z)| \le 3^s 3^2 \le 27,$$

whereas on the real interval,

$$F_{\mathrm{mid}}(v) \ge (4/3)^{-s-2} \ge \frac{27}{64}.$$

Equation (19) gives relative error at most $256\, 4^{-r}$ on every middle panel.

For the high tail, the substitution $\tau = C/u$ gives

$$c_s \int_C^{\infty} \frac{\tau^{-s}}{(x+\tau)^2}\, d\tau = c_s C^{1-s} \int_0^1 \frac{u^s}{(C+xu)^2}\, du.$$

The Gaussian rule in (17) is for $u^{s-1}\, du$. Its derivative integrand therefore contains an additional factor u. With $v = 2u - 1$, the kernel, apart from $(C + x/2)^{-2}$, is

$$\ell(v) F_{\mathrm{hi}}(v), \quad \ell(v) = \frac{1+v}{2}, \quad F_{\mathrm{hi}}(v) = (1+\theta v)^{-2},$$

where $\theta = x/(2C+x) \le 1/3$, since $x \le b \le C$. The same bounds $M = 9$ and $m = 9/16$ hold. Equation (20) gives the high-tail relative error $128\, 4^{-r}$. This rule is precisely the high-tail contribution to D_p, because

$$\frac{\omega_i^{\mathrm{hi}}}{(x+\tau_i^{\mathrm{hi}})^2} = c_s C^{1-s} B_i \frac{v_i}{(C + x v_i)^2}.$$

Let $I_\nu(x) > 0$ be the exact derivative contribution of a panel, and let $\widetilde{I}_\nu(x)$ be its quadrature. Each preceding bound is at most $256\, 4^{-r} I_\nu(x)$. Thus

$$\begin{aligned} |D_p(x) - D(x)| &\le \sum_\nu |\widetilde{I}_\nu(x) - I_\nu(x)| \\ &\le 256\, 4^{-r} \sum_\nu I_\nu(x) = 256\, 4^{-r} D(x). \end{aligned}$$

Positivity prevents a factor proportional to the number of panels from appearing. □

Theorem 1 (Dimension-free relative marginal approximation)**.** *Let $M \succ 0$ and $b_e \neq 0$, and suppose that*

$$\operatorname{spec}(M + tb_e b_e^\top) \subseteq [a, b] \quad \textit{for every } t \in [0, 1].$$

Let $0 < s < 1$ and $0 < \epsilon \leq 1/2$. Construct (18) *with $r = \lceil \log_4(256/\epsilon) \rceil$. All weights and poles are positive, and*

$$p = \big(\lceil \log_2(b/a) \rceil + 2\big) r = O\big((1 + \log(b/a)) \log(1/\epsilon)\big).$$

The rational marginal is

$$\widetilde{\Delta}_s(e; M) = -\sum_{j=1}^{p} \omega_j \frac{b_e^\top (M + \tau_j I)^{-2} b_e}{1 + b_e^\top (M + \tau_j I)^{-1} b_e}. \tag{23}$$

For $\Delta_s(e; M) = \operatorname{Tr}((M + b_e b_e^\top)^{-s}) - \operatorname{Tr}(M^{-s})$, it satisfies

$$|\widetilde{\Delta}_s(e; M) - \Delta_s(e; M)| \leq \epsilon |\Delta_s(e; M)|. \tag{24}$$

Proof. Equations (15)–(17) give positive weights and poles, numbering $(K + 2)r$, which proves the pole-count assertion.

Write $R_j = (M + \tau_j I)^{-1}$. Its positive definiteness implies $1 + b_e^\top R_j b_e > 0$. The Sherman–Morrison identity gives

$$(M + \tau_j I + b_e b_e^\top)^{-1} = R_j - \frac{R_j b_e b_e^\top R_j}{1 + b_e^\top R_j b_e}.$$

Taking traces and using their cyclic invariance yields

$$\operatorname{Tr}\big((M + \tau_j I + b_e b_e^\top)^{-1} - R_j\big) = -\frac{b_e^\top R_j^2 b_e}{1 + b_e^\top R_j b_e}.$$

Multiplication by ω_j and summation proves (23); it is exactly the trace difference of r_p.

Set $M_t = M + tb_e b_e^\top$. Differentiating the resolvent gives

$$\frac{d}{dt}(M_t + \tau I)^{-1} = -(M_t + \tau I)^{-1} b_e b_e^\top (M_t + \tau I)^{-1}.$$

The absolute value of its trace is at most $\|b_e\|^2/(a+\tau)^2$. Since $\int_0^\infty \tau^{-s}/(a + \tau)^2 \, d\tau$ is finite, differentiation under the trace Stieltjes integral is justified. Consequently,

$$\begin{aligned} \frac{d}{dt} \operatorname{Tr}(M_t^{-s}) &= -c_s \int_0^\infty \tau^{-s} b_e^\top (M_t + \tau I)^{-2} b_e \, d\tau \\ &= -b_e^\top D(M_t) b_e. \end{aligned}$$

The last equality follows by applying (22) to each eigenvalue of M_t. Differentiating the finite rational sum similarly gives

$$\frac{d}{dt} \operatorname{Tr}(r_p(M_t)) = -b_e^\top D_p(M_t) b_e.$$

The fundamental theorem of calculus now yields

$$-\Delta_s(e; M) = \int_0^1 b_e^\top D(M_t) b_e \, dt,$$

and

$$-\widetilde{\Delta}_s(e; M) = \int_0^1 b_e^\top D_p(M_t) b_e \, dt.$$

Lemma 3 and the chosen r imply

$$(1 - \epsilon) D(x) \leq D_p(x) \leq (1 + \epsilon) D(x) \quad (x \in [a, b]).$$

Diagonalize $M_t = V_t \Lambda_t V_t^\top$. Both $D(M_t)$ and $D_p(M_t)$ use this same eigenbasis, and every eigenvalue belongs to $[a, b]$. Applying the scalar inequalities to each diagonal entry therefore proves

$$(1 - \epsilon) D(M_t) \preceq D_p(M_t) \preceq (1 + \epsilon) D(M_t).$$

Taking quadratic forms with b_e and integrating gives

$$(1 - \epsilon)(-\Delta_s(e; M)) \leq -\widetilde{\Delta}_s(e; M) \leq (1 + \epsilon)(-\Delta_s(e; M)).$$

Since $D(M_t) = sM_t^{-s-1} \succ 0$ and $b_e \neq 0$, the quantity $-\Delta_s(e; M)$ is positive. Rearranging proves (24). □

For unit candidate edges, $\|b_e\|^2 = 2$. Every edge-addition path ending in a set of at most $2k$ edges is therefore covered by

$$a = \sigma, \qquad b = \Lambda = \sigma + 2d_{\max,0} + 4k, \qquad \kappa_* = \Lambda/\sigma.$$

The estimate follows from $\|L_0\| \leq 2d_{\max,0}$ and $\|\sum_{e \in S} b_e b_e^\top\| \leq 2|S|$. The marginal guarantee compares positive quadratic-form integrals along the update path; it does not subtract two separately approximated n-dimensional traces.

For algebraic re-ranking, set $u_j = (M + \tau_j I)^{-1} b_e$. Symmetry gives $b_e^\top (M + \tau_j I)^{-2} b_e = u_j^\top u_j$. Thus the positive rational marginal is

$$G_p(e \mid S) = -\widetilde{\Delta}_s(e; M_S) = \sum_{j=1}^{p} \omega_j \frac{\|u_j\|_2^2}{1 + b_e^\top u_j}.$$

It requires no random projection once these solves are available. Its remaining difference from the true fractional marginal is the quadrature error in Theorem 1.

C. The Log-Trace Reference Objective

Proposition 2 (Strict monotonicity and submodularity of log–trace)**.** *Write $g_{\log}(e \mid S) = f_{\log}(S \cup \{e\}) - f_{\log}(S)$. The normalized log-trace gain $f_{\log}(S) = \log\det M(S) - \log\det M(\varnothing)$ is strictly increasing under every nonzero edge addition and is submodular. For two candidates, strict diminishing returns holds exactly when their inverse-matrix cross term is nonzero.*

Proof. Fix S and an unselected edge e, and write $R = M(S)^{-1}$. The matrix determinant lemma gives

$$g_{\log}(e \mid S) = \log(1 + b_e^\top R b_e) > 0,$$

because R is positive definite and $b_e \neq 0$. For $S \subseteq T$, let $A = M(S)$ and $H = M(T) - M(S) \succeq 0$. Writing $C = A^{-1/2} H A^{-1/2} \succeq 0$ gives

$$M(T)^{-1} = A^{-1/2}(I + C)^{-1} A^{-1/2} \preceq A^{-1},$$

since every eigenvalue of $(I + C)^{-1}$ is at most one. Thus $b_e^\top M(T)^{-1} b_e \leq b_e^\top M(S)^{-1} b_e$; applying the increasing function $x \mapsto \log(1 + x)$ proves diminishing returns.

More explicitly, for distinct unselected edges e, a, put $d_e = b_e^\top R b_e$, $d_a = b_a^\top R b_a$, and $c = b_e^\top R b_a$. Sherman–Morrison gives

$$\begin{aligned} &g_{\log}(e \mid S \cup \{a\}) - g_{\log}(e \mid S) \\ &\qquad = \log\left(1 - \frac{c^2}{(1 + d_e)(1 + d_a)}\right) \leq 0. \end{aligned}$$

The logarithm is well-defined because Cauchy–Schwarz in the R-inner product yields $c^2 \le d_e d_a < (1+d_e)(1+d_a)$. The inequality is strict exactly when $c \neq 0$.

For a connected equality example, take the unweighted four-cycle $E_0 = \{12, 23, 34, 14\}$ and its two missing diagonals $e = 13$ and $a = 24$. Their incidence vectors are orthogonal eigenvectors of L_0, both with eigenvalue 2. Hence $Rb_a = b_a/(\sigma+2)$ and $b_e^\top R b_a = 0$. The preceding equality shows that adding one diagonal does not change the log-trace marginal of the other. □

D. Submodularity Ratio and Budget-Restricted Curvature

Write

$$g_s(e \mid S) = f_s(S \cup \{e\}) - f_s(S),$$
$$g_s(T \mid S) = f_s(S \cup T) - f_s(S).$$

For a budget $k \ge 1$, define the restricted submodularity ratio by

$$\gamma_k = \inf_{\substack{S \cap T = \varnothing,\ |S| \le k-1 \\ 1 \le |T| \le k}} \frac{\sum_{e \in T} g_s(e \mid S)}{g_s(T \mid S)}. \tag{25}$$

The restricted generalized curvature is

$$\alpha_{2k} = 1 - \inf_{\substack{A \subseteq B,\ e \notin B \\ |B|+1 \le 2k}} \frac{g_s(e \mid B)}{g_s(e \mid A)}. \tag{26}$$

All sets in these definitions are subsets of $\mathcal{C}$. The strict positivity in Lemma 1 makes every displayed ratio well-defined. Including singleton T and $A = B$ shows that $\gamma_k \le 1$ and $\alpha_{2k} \ge 0$. The $2k$ restriction covers the union of a greedy solution and an optimal budget-k set.

Theorem 2 (Submodularity ratio and curvature)**.** *Let $0 < s < 1$, $\sigma > 0$, and let each candidate be a unit-weight undirected edge, so that $\|b_e\|_2^2 = 2$. Let $d_{\max,0}$ denote the maximum weighted degree of L_0, and put*

$$\Lambda = \sigma + 2d_{\max,0} + 4k. \tag{27}$$

Then f_s is normalized and strictly increasing, and

$$\gamma_k \ge \gamma_0 := \frac{\sigma}{2s}\left[1 - \left(\frac{\sigma}{\sigma+2}\right)^s\right] > 0, \tag{28}$$

$$\alpha_{2k} \le \alpha_0 := 1 - \left(\frac{\sigma}{\Lambda}\right)^{s+1} < 1. \tag{29}$$

In fact, the bound in (28) *holds without a cardinality restriction on S or T.*

Proof. Set $c_s = \sin(\pi s)/\pi > 0$. Fix an SPD matrix $A \succeq \sigma I$ and a nonzero unit-edge vector b. For $\tau > 0$, write $R_\tau = (A + \tau I)^{-1}$, $d_\tau = b^\top R_\tau b$, and $h_\tau = b^\top R_\tau^2 b$. The Stieltjes representation and the Sherman–Morrison identity give

$$\operatorname{Tr}(A^{-s}) - \operatorname{Tr}((A + bb^\top)^{-s}) = c_s \int_0^\infty \tau^{-s} \frac{h_\tau}{1 + d_\tau}\, d\tau. \tag{30}$$

Every integrand is strictly positive. The integral is finite because it is $O(\tau^{-s})$ at zero and $O(\tau^{-s-2})$ at infinity. Consequently every candidate has a strictly positive marginal, and $f_s(\varnothing) = 0$ proves normalization and strict monotonicity.

Differentiating the scalar Stieltjes identity under the integral gives

$$s x^{-s-1} = c_s \int_0^\infty \frac{\tau^{-s}}{(x+\tau)^2}\, d\tau.$$

This differentiation is justified locally uniformly for $x > 0$ by an integrable bound of the form $C\tau^{-s}$ on $(0,1)$ and $C\tau^{-s-2}$ on $(1,\infty)$. Applying the resulting identity in an eigenbasis of A yields

$$D(A,b) := s b^\top A^{-s-1} b = c_s \int_0^\infty \tau^{-s} h_\tau\, d\tau. \tag{31}$$

Since $d_\tau \le \|b\|_2^2/(\sigma+\tau) = 2/(\sigma+\tau)$,

$$\operatorname{Tr}(A^{-s}) - \operatorname{Tr}((A+bb^\top)^{-s}) \ge c_s \int_0^\infty \tau^{-s} h_\tau \frac{\sigma+\tau}{\sigma+\tau+2}\, d\tau. \tag{32}$$

Let $\mu_i \ge \sigma$ be the eigenvalues of A, with orthonormal eigenvectors v_i, and let $z_i = (v_i^\top b)^2 \ge 0$. Then $h_\tau = \sum_i z_i/(\mu_i+\tau)^2$. For $\mu \ge \sigma$, define

$$I(\mu) = \int_0^\infty \frac{\tau^{-s}}{(\mu+\tau)^2} \frac{\sigma+\tau}{\sigma+\tau+2}\, d\tau.$$

The substitution $\tau = \mu u$ gives

$$\begin{aligned} I(\mu) &= \mu^{-s-1} \int_0^\infty \frac{u^{-s}}{(1+u)^2} \frac{\sigma+\mu u}{\sigma+\mu u+2}\, du \\ &\ge \left(\frac{\mu}{\sigma}\right)^{-s-1} I(\sigma). \end{aligned} \tag{33}$$

Indeed, $y/(y+2)$ is increasing for $y > 0$, and $\mu \ge \sigma$. At the endpoint $\mu = \sigma$, partial fractions and the Stieltjes identity imply

$$\begin{aligned} c_s I(\sigma) &= c_s \int_0^\infty \frac{\tau^{-s}}{(\sigma+\tau)(\sigma+\tau+2)}\, d\tau \\ &= \frac{1}{2}\left[\sigma^{-s} - (\sigma+2)^{-s}\right] = \gamma_0 s \sigma^{-s-1}. \end{aligned}$$

Substituting this equality into (33) and then into (32) proves

$$\operatorname{Tr}(A^{-s}) - \operatorname{Tr}((A+bb^\top)^{-s}) \ge \gamma_0 s \sum_i z_i \mu_i^{-s-1} = \gamma_0 D(A,b). \tag{34}$$

To compare this single-edge inequality with a joint gain, let $H = \sum_{e \in T} b_e b_e^\top \succeq 0$ and $K_\tau = R_\tau^{1/2} H R_\tau^{1/2} \succeq 0$. Since $x/(1+x) \le x$ for $x \ge 0$, the spectral theorem gives

$$0 \preceq I - (I + K_\tau)^{-1} \preceq K_\tau.$$

Congruence with $R_\tau^{1/2}$ therefore gives

$$\begin{aligned} R_\tau - (A + H + \tau I)^{-1} &= R_\tau^{1/2}[I - (I+K_\tau)^{-1}] R_\tau^{1/2} \\ &\preceq R_\tau H R_\tau. \end{aligned}$$

Taking traces, integrating with positive weight $c_s \tau^{-s}$, and using (31) yields

$$\operatorname{Tr}(A^{-s}) - \operatorname{Tr}((A+H)^{-s}) \le \sum_{e \in T} D(A, b_e). \tag{35}$$

Apply (34) to each $e \in T$ at $A = M(S)$ and sum. Combining the result with (35) proves $\sum_{e\in T} g_s(e \mid S) \geq \gamma_0 g_s(T \mid S)$. No cardinality restriction was used in this argument. The identity

$$\gamma_0 = \int_0^1 \left(\frac{\sigma}{\sigma + 2t}\right)^{s+1} dt$$

also shows directly that $0 < \gamma_0 < 1$.

It remains to prove the curvature bound. For every vector x,

$$\begin{aligned} x^\top L_0 x &= \sum_{\{u,v\}\in E_0} w_{uv}(x_u - x_v)^2 \\ &\leq 2\sum_v d_{v,0} x_v^2 \leq 2d_{\max,0}\|x\|_2^2. \end{aligned}$$

Each added edge has operator norm $\|b_e b_e^\top\|_2 = 2$. Thus every matrix formed from at most $2k$ added unit edges, including any partially added last edge, has all eigenvalues in $[\sigma, \Lambda]$. The resolvent derivative $\frac{d}{dt}(A + tbb^\top + \tau I)^{-1} = -(A + tbb^\top + \tau I)^{-1} bb^\top (A + tbb^\top + \tau I)^{-1}$, the same integrable bounds as above, and cyclicity of the trace give

$$g_s(e \mid S) = s\int_0^1 b_e^\top (M(S) + tb_e b_e^\top)^{-s-1} b_e \, dt. \tag{36}$$

The spectral theorem, applied separately at each t, implies

$$\Lambda^{-s-1} I \preceq (M(S) + tb_e b_e^\top)^{-s-1} \preceq \sigma^{-s-1} I.$$

Hence whenever $|S| + 1 \leq 2k$,

$$2s\Lambda^{-s-1} \leq g_s(e \mid S) \leq 2s\sigma^{-s-1}.$$

For every pair $A \subseteq B$ appearing in (26), these bounds give

$$\frac{g_s(e \mid B)}{g_s(e \mid A)} \geq \left(\frac{\sigma}{\Lambda}\right)^{s+1}.$$

Taking the infimum proves (29). The comparison applies scalar spectral bounds separately to each path matrix. □

E. A Complete Approximate-Greedy Guarantee

The following proof gives the error-sensitive form of the ratio–curvature guarantee [10] using only the restricted parameters above.

Lemma 4 (Approximate greedy selection)**.** *Let f_s be normalized and monotone, with the ratios in* (25)–(26) *well-defined. Suppose that $\gamma_k \geq \gamma > 0$ and $\alpha_{2k} \leq \alpha \leq 1$ under* (25)–(26)*. Let $S_0 = \varnothing$ and $S_i = S_{i-1} \cup \{e_i\}$, where*

$$g_s(e_i \mid S_{i-1}) \geq \beta \max_{e\in\mathcal{C}\setminus S_{i-1}} g_s(e \mid S_{i-1}), \qquad 0 < \beta \leq 1. \tag{37}$$

For an optimal set O of cardinality at most k, put $\theta = \beta\gamma$ and

$$H_k(\alpha, \theta) := \frac{\theta}{k}\sum_{h=0}^{k-1}\left(1 - \frac{\alpha\theta}{k}\right)^h. \tag{38}$$

Then $f_s(S_k) \geq H_k(\alpha, \theta) f_s(O)$. For $\alpha > 0$,

$$H_k(\alpha, \theta) = \frac{1 - (1 - \alpha\theta/k)^k}{\alpha} \geq \frac{1 - e^{-\alpha\theta}}{\alpha};$$

for $\alpha = 0$, $H_k(0, \theta) = \theta$.

Proof. If $f_s(O) = 0$ the claim is immediate; otherwise extend O to size k by monotonicity. Write $\rho_i = g_s(e_i \mid S_{i-1})$ and $w_t = |S_t \cap O|$. Adding $S_t \setminus O$ to O in greedy order and applying curvature gives

$$f_s(O \cup S_t) - f_s(O) \geq (1 - \alpha) \sum_{i\leq t : e_i \notin O} \rho_i.$$

The ratio inequality and β-greedy selection give

$$f_s(O \cup S_t) - f_s(S_t) \leq \frac{k - w_t}{\beta\gamma}\rho_{t+1}.$$

Only unions of at most $2k$ edges occur. Setting $x_i = \rho_i / f_s(O)$ and $\theta = \beta\gamma$ yields

$$1 \leq \alpha \sum_{i\leq t : e_i \notin O} x_i + \sum_{i\leq t : e_i \in O} x_i + \frac{k - w_t}{\theta} x_{t+1}.$$

For completeness, solve the resulting scalar relaxation. At state (m, R, z), $m \leq R$ selections remain and $x \geq \theta z / R$ is required. An outside selection reduces z by αx and leaves R unchanged; an inside selection reduces z by x and R by one. Positive parts are taken after each reduction. Write the minimum cost as $zC_{m,R}$, with $C_{0,R} = 0$. We have $0 \leq C_{m,R} \leq 1$, since choosing an inside increment $x = z$ extinguishes the residual. Hence $x + C_{\text{next}}[z - cx]_+$ is nondecreasing in x for $c \in \{\alpha, 1\}$, and

$$\begin{aligned} C_{m,R} = \frac{\theta}{R} + \min\Big\{&(1 - \alpha\theta/R) C_{m-1,R}, \\ &(1 - \theta/R) C_{m-1,R-1}\Big\}. \end{aligned}$$

Induction gives, for $1 \leq m \leq R$, $C_{m,R} = (\theta/R)\sum_{h=0}^{m-1}(1 - a/R)^h$, where $a = \alpha\theta \in [0, 1]$. Indeed, the outside branch gives this expression, while the inside branch is no smaller because, for $h \leq m - 2 \leq R - 2$,

$$\frac{R - \theta}{R - 1}(1 - a/(R - 1))^h \geq (1 - a/R)^{h+1}.$$

For $a = 0$ or $R = 2$ this is immediate. Otherwise $J(u) = (1 - a/u)^{u-1}$ is nonincreasing, since

$$\begin{aligned} (\log J)'(u) &= \log(1 - a/u) + \frac{a(u-1)}{u(u-a)} \\ &\leq \log(1 - a/u) + a/u \leq 0. \end{aligned}$$

The ratio $(1 - a/R)^{h+1}/(1 - a/(R-1))^h$ increases with h and is at most $J(R)/J(R-1) \leq 1$. This proves the induction. The initial state $(k, k, 1)$ gives $\sum_i x_i \geq C_{k,k} = H_k(\alpha, \theta)$. Summing the geometric series and using $(1 - \alpha\theta/k)^k \leq e^{-\alpha\theta}$ proves the claim, including its continuous limit at $\alpha = 0$. □

F. Numerical Errors and Algebraic Re-ranking

Lanczos matrix-function approximation admits finite-precision stability and a posteriori error analyses [18], [19]. For our marginal oracle, the following residual certificate separates linear-solve error from the quadrature error in Theorem 1.

Lemma 5 (Residual certification of algebraic reranking scores)**.** *Fix a state S, a candidate e, and positive quadrature weights*

ω_j. *Set* $A_j = M(S) + \tau_j I$ *and* $u_j = A_j^{-1} b_e$. *Assume* $a_j I \preceq A_j \preceq b_j I$, *where one may take* $a_j = \sigma + \tau_j$ *and* $b_j = \Lambda + \tau_j$. *For some* $0 < \rho < 1$, *suppose the computed vectors satisfy*

$$\|A_j \widetilde{u}_j - b_e\|_2 \le \rho \frac{a_j}{b_j} \|b_e\|_2 \qquad (j = 1, \dots, p). \tag{39}$$

Define the exact rational and computed scores by

$$G_p(e \mid S) = \sum_{j=1}^{p} \omega_j \frac{\|u_j\|_2^2}{1 + b_e^\top u_j},$$
$$\check{G}(e \mid S) = \sum_{j=1}^{p} \omega_j \frac{\|\widetilde{u}_j\|_2^2}{1 + b_e^\top \widetilde{u}_j}.$$

Then every computed denominator is positive and

$$\frac{(1-\rho)^2}{1+\rho} G_p(e \mid S) \le \check{G}(e \mid S) \le \frac{(1+\rho)^2}{1-\rho} G_p(e \mid S). \tag{40}$$

In particular, if $0 < \varepsilon_r \le 1/2$ *and* $\rho \le \varepsilon_r/4$, *then* $(1-\varepsilon_r)G_p \le \check{G} \le (1+\varepsilon_r)G_p$.

Proof. Write $r_j = A_j \widetilde{u}_j - b_e$ and $v_j = \widetilde{u}_j - u_j = A_j^{-1} r_j$. The assumed residual bound gives

$$\|v_j\|_2 \le a_j^{-1} \|r_j\|_2 \le \rho \frac{\|b_e\|_2}{b_j}.$$

Also, $\|b_e\|_2 = \|A_j u_j\|_2 \le b_j \|u_j\|_2$ and $d_j := b_e^\top u_j \ge \|b_e\|_2^2 / b_j > 0$. Consequently

$$\|v_j\|_2 \le \rho \|u_j\|_2, \qquad |b_e^\top v_j| \le \|b_e\|_2 \|v_j\|_2 \le \rho d_j.$$

The triangle and reverse triangle inequalities imply

$$(1-\rho)^2 \|u_j\|_2^2 \le \|\widetilde{u}_j\|_2^2 \le (1+\rho)^2 \|u_j\|_2^2.$$

For the denominator,

$$1 + b_e^\top \widetilde{u}_j \ge 1 + (1-\rho) d_j \ge (1-\rho)(1+d_j) > 0,$$
$$1 + b_e^\top \widetilde{u}_j \le 1 + (1+\rho) d_j \le (1+\rho)(1+d_j).$$

Combining these inequalities proves (40) for each summand; positivity of the weights preserves the bounds after summation. Finally, $\rho \le \varepsilon_r/4 \le 1/8$ implies

$$\frac{(1+\rho)^2}{1-\rho} - 1 = \frac{3\rho + \rho^2}{1-\rho} \le 4\rho \le \varepsilon_r,$$

where $\rho \le 1/5$ suffices for the middle inequality, and

$$1 - \frac{(1-\rho)^2}{1+\rho} = \frac{3\rho - \rho^2}{1+\rho} \le 3\rho \le \varepsilon_r.$$

These inequalities prove the last assertion. □

Theorem 3 (End-to-end guarantee with certified errors)**.** *Let* G_p *be the positive multi-pole marginal, and suppose quadrature satisfies, uniformly over all states of size less than* k *and their remaining candidates,*

$$(1-\varepsilon_Q) g_s(e \mid S) \le G_p(e \mid S) \le (1+\varepsilon_Q) g_s(e \mid S). \tag{41}$$

Suppose that on an event $\mathcal{E}$, *the coarse scores satisfy*

$$(1-\varepsilon_c) G_p(e \mid S) \le \widehat{G}(e \mid S) \le (1+\varepsilon_c) G_p(e \mid S) \tag{42}$$

for all candidates at every state actually visited. At each round, let the shortlist contain a global maximizer of $\widehat{G}$, *and select from it a maximizer of a reranking score satisfying*

$$(1-\varepsilon_r) G_p(e \mid S) \le \check{G}(e \mid S) \le (1+\varepsilon_r) G_p(e \mid S). \tag{43}$$

Assume $0 \le \varepsilon_Q, \varepsilon_c, \varepsilon_r < 1$, *and put*

$$\beta = \frac{1-\varepsilon_Q}{1+\varepsilon_Q} \frac{1-\varepsilon_c}{1+\varepsilon_c} \frac{1-\varepsilon_r}{1+\varepsilon_r}.$$

Then, on $\mathcal{E}$, *the resulting set satisfies*

$$f_s(S_k) \ge H_k(\alpha_0, \beta\gamma_0) f_s(O) \ge \frac{1 - e^{-\alpha_0 \beta \gamma_0}}{\alpha_0} f_s(O), \tag{44}$$

where O *is optimal under the budget* k *and* γ_0, α_0 *are given in* (28)–(29). *If* $\Pr(\mathcal{E}) \ge 1 - \delta$, *the same guarantee holds with probability at least* $1 - \delta$.

Proof. Work on $\mathcal{E}$ and fix a visited state S. Let c be a global maximizer of $\widehat{G}(\cdot \mid S)$ contained in the shortlist, let a be the reranking winner, and let e be any remaining candidate. Suppress S in the following display. Successive applications of the three error bounds and the two maximization rules give

$$\begin{aligned}
g_s(a) &\ge \frac{G_p(a)}{1+\varepsilon_Q} \\
&\ge \frac{\check{G}(a)}{(1+\varepsilon_Q)(1+\varepsilon_r)} \\
&\ge \frac{\check{G}(c)}{(1+\varepsilon_Q)(1+\varepsilon_r)} \\
&\ge \frac{(1-\varepsilon_r) G_p(c)}{(1+\varepsilon_Q)(1+\varepsilon_r)} \\
&\ge \frac{(1-\varepsilon_r) \widehat{G}(c)}{(1+\varepsilon_Q)(1+\varepsilon_r)(1+\varepsilon_c)} \\
&\ge \frac{(1-\varepsilon_r) \widehat{G}(e)}{(1+\varepsilon_Q)(1+\varepsilon_r)(1+\varepsilon_c)} \\
&\ge \frac{(1-\varepsilon_r)(1-\varepsilon_c) G_p(e)}{(1+\varepsilon_Q)(1+\varepsilon_r)(1+\varepsilon_c)} \\
&\ge \beta g_s(e).
\end{aligned}$$

Thus the selected edge satisfies (37) at every round. Theorem 2 supplies the inequalities required by Lemma 4 with $\gamma = \gamma_0$ and $\alpha = \alpha_0$. Applying that lemma proves (44). The event implication immediately gives the probability statement. □

The factor β separates quadrature, maintained-proxy, and algebraic-score errors. Lemma 5 calibrates solve tolerances; the next corollary calibrates fixed projections. For exact true-marginal greedy, $\beta = 1$ and the factor becomes $(1 - e^{-\alpha_0 \gamma_0})/\alpha_0$.

Remark 1 (Round-dependent recovery)**.** The factor β is a worst case over rounds. If round i satisfies (37) with its own β_i, the standard ratio argument gives $f_s(O) - f_s(S_i) \le (1 - \gamma_k \beta_i / k)(f_s(O) - f_s(S_{i-1}))$, hence $f_s(S_k) \ge \big(1 - e^{-\gamma_k \bar{\beta}}\big) f_s(O)$ with $\bar{\beta} = k^{-1} \sum_i \beta_i$. Section V-F reports both the minimum and the mean of the measured per-round recovery.

G. A Uniform Certificate for Fixed Sketches

The following simultaneous guarantee makes the quantifiers needed for adaptive selection explicit; the underlying sketch constructions are standard [20], [21].

Corollary 1 (Uniform fixed-sketch guarantee)**.** *Fix $0 < \eta < 1/3$, $0 < \delta < 1$, $1 \le k \le N = |\mathcal{C}|$, and positive quadrature weights and shifts $(\omega_j, \tau_j)_{j=1}^{p}$. Let m_0 be the number of base edges, and let $C_S \in \mathbb{R}^{(m_0+N)\times n}$ contain their weighted incidence rows $\sqrt{w_a} b_a^\top$, followed by the incidence row of each selected unit-weight candidate and a zero row for each unselected candidate. Thus $C_S^\top C_S = M(S) - \sigma I$. For $|S| < k$, $e \in \mathcal{C} \setminus S$, and $j \in [p]$, define*

$$R_{S,j} = (M(S) + \tau_j I)^{-1}, \quad v_{S,e,j} = R_{S,j} b_e,$$

$$h_{S,e,j} = \|v_{S,e,j}\|^2,$$
$$t_{S,e,j} = \|C_S v_{S,e,j}\|^2,$$
$$d_{S,e,j} = b_e^\top R_{S,j} b_e.$$

Draw $Z \in \mathbb{R}^{n\times q}$ and $U \in \mathbb{R}^{(m_0+N)\times q}$ independently, with independent $N(0, 1/q)$ entries, and keep them fixed. Put $\widehat{h}_{S,e,j} = \|Z^\top v_{S,e,j}\|^2$ and $\widehat{t}_{S,e,j} = \|U^\top C_S v_{S,e,j}\|^2$. The exact rational gain and its sketch are

$$G_p(e \mid S) = \sum_{j=1}^{p} \omega_j \frac{h_{S,e,j}}{1 + d_{S,e,j}}, \tag{45}$$

$$\widehat{G}(e \mid S) = \sum_{j=1}^{p} \omega_j \frac{\widehat{h}_{S,e,j}}{1 + \widehat{t}_{S,e,j} + (\sigma + \tau_j)\widehat{h}_{S,e,j}}. \tag{46}$$

Let $\mathcal{N}_k = \sum_{\ell=0}^{k-1} \binom{N}{\ell}$. If

$$q \ge \frac{8}{\eta^2} \log \frac{4pN\mathcal{N}_k}{\delta}, \tag{47}$$

then, with probability at least $1 - \delta$, simultaneously for every such (S, e),

$$\frac{1-\eta}{1+\eta} G_p(e \mid S) \le \widehat{G}(e \mid S) \le \frac{1+\eta}{1-\eta} G_p(e \mid S). \tag{48}$$

In particular, this event covers every trajectory of at most k adaptive selections using these same sketches. It also implies a conventional relative-error bound with $\varepsilon_c = 2\eta/(1-\eta) < 1$.

Proof. For a fixed nonzero vector x and Gaussian W, $q\|W^\top x\|^2/\|x\|^2$ is χ_q^2. The moment generating function $\mathbb{E}e^{t\chi_q^2} = (1-2t)^{-q/2}$ and Chernoff optimization give, for $0 < a < 1$,

$$\Pr\{\chi_q^2 \ge q(1+a)\} \le e^{-q(a - \log(1+a))/2},$$

and $\Pr\{\chi_q^2 \le q(1-a)\} \le e^{-q(-a - \log(1-a))/2}$. Since $a - \log(1+a) \ge a^2/4$ and $-a - \log(1-a) \ge a^2/2$, the squared-norm failure probability is at most $2e^{-qa^2/8}$. Zero vectors are preserved exactly.

There are at most $2pN\mathcal{N}_k$ deterministic norm queries over all (S, e, j) before the sketches are drawn. A union bound with $a = \eta$ gives failure probability at most $4pN\mathcal{N}_k e^{-q\eta^2/8} \le \delta$. On the complementary event, both h and t are preserved within $1 \pm \eta$. Moreover,

$$t + (\sigma + \tau_j)h = v^\top (M_S + \tau_j I) v = v^\top b_e = d.$$

Thus $(1-\eta)(1+d) \le 1 + \widehat{t} + (\sigma + \tau_j)\widehat{h} \le (1+\eta)(1+d)$. Dividing the numerator bounds by these positive denominator bounds and summing with $\omega_j > 0$ proves (48). Its upper factor is $1 + \varepsilon_c$ and its lower factor is at least $1 - \varepsilon_c$. The event covers all admissible sets simultaneously, hence also every adaptive trajectory. □

For $N \ge 2$,

$$\mathcal{N}_k \le \sum_{\ell=0}^{k-1} N^\ell = \frac{N^k - 1}{N - 1} \le N^k;$$

for $N = 1$, necessarily $k = 1$ and $\mathcal{N}_k = 1$. Thus a sufficient Gaussian dimension is $O(\eta^{-2}[(k+1)\log N + \log(p/\delta)])$. This bound accounts for all possible adaptive trajectories, not only a trajectory fixed independently of the sketches.

Corollary 2 (Global orthogonal-column variant)**.** *In Corollary 1, replace the two Gaussian sketches by independent scaled Haar frames $\sqrt{d_i/q}\, Q_i$, where the two Q_i have q orthonormal columns in dimensions $d_1 = n$ and $d_2 = m_0 + N$, respectively. For $q \le \min(d_1, d_2)$, the same conclusion holds under the sufficient condition*

$$q \ge \frac{72}{\eta^2} \log \frac{8pN\mathcal{N}_k}{\delta}. \tag{49}$$

The node sketch may instead be a Haar frame in $\mathbf{1}^\perp$, scaled by $\sqrt{(n-1)/q}$, with $d_1 = n - 1$.

Proof. Rotational invariance gives, for a fixed nonzero $x \in \mathbb{R}^d$,

$$\frac{\|\sqrt{d/q}\, Q^\top x\|^2}{\|x\|^2} \overset{\mathrm{d}}{=} \frac{A/q}{(A+B)/d},$$

where $A \sim \chi_q^2$ and $B \sim \chi_{d-q}^2$ are independent. Apply the preceding chi-square bound to A and $A + B$ with tolerance $\eta/3$. When both normalized variables lie in $1 \pm \eta/3$, their ratio lies in $1 \pm \eta$, since $2\eta/(3-\eta) \le \eta$ and $2\eta/(3+\eta) \le \eta$. The fixed-vector failure probability is therefore at most $4e^{-q\eta^2/72}$ for $q \le d$. Union bounding the $2pN\mathcal{N}_k$ queries proves the stated dimension bound; the score comparison follows from the same positive-denominator argument.

For centered node sketches, $\mathbf{1}^\top R_{S,j} b_e = \mathbf{1}^\top b_e/(\sigma + \tau_j) = 0$. Every projected node vector therefore lies in $\mathbf{1}^\perp$, where the preceding argument applies with dimension $n - 1$. □

Corollary 2 applies when its lower bound on q is compatible with the ambient dimension; a full-dimensional isometry gives exact norm preservation. Numerical maintenance errors enter ε_c through the residual bounds in Section IV-B.

IV. The POLAR Framework

We maintain a fixed two-bank sketch and select a proxy shortlist before algebraic re-ranking. The candidate support $V_{\mathcal{C}}$ is fixed throughout.

A. Shared Krylov Iterations and Full-Norm Tracking

Sort the poles so that $\tau_1 \le \cdots \le \tau_p$. For a common right-hand side w, use $A = M + \tau_1 I$ as the seed matrix and $\theta_j = \tau_j - \tau_1$. Write u_i, r_i, d_i for its CG iterate, residual, and search direction, initialized by $u_0 = 0$, $r_0 = d_0 = w$. Every shift starts from $u_0^{(j)} = 0$ and $d_0^{(j)} = w$; $w = 0$ is returned immediately. One seed iteration is

$$a_i = \frac{r_i^\top r_i}{d_i^\top A d_i}, \qquad r_{i+1} = r_i - a_i A d_i, \tag{50}$$

$$\beta_i = \frac{\|r_{i+1}\|^2}{\|r_i\|^2}, \qquad d_{i+1} = r_{i+1} + \beta_i d_i. \tag{51}$$

Only $Ad_i = Md_i + \tau_1 d_i$ requires a graph SpMV. Set $\zeta_{-1}^{(j)} = \zeta_0^{(j)} = 1$ and $\chi_0 = 0$, with $\chi_i = a_i\beta_{i-1}/a_{i-1}$ for $i > 0$. The coupled scalar recurrences [9] are

$$\zeta_{i+1}^{(j)} = \frac{\zeta_i^{(j)}}{1 + a_i\theta_j + \chi_i(1 - \zeta_i^{(j)}/\zeta_{i-1}^{(j)})}, \tag{52}$$

$$a_i^{(j)} = a_i \frac{\zeta_{i+1}^{(j)}}{\zeta_i^{(j)}}, \tag{53}$$

$$\beta_i^{(j)} = \beta_i \left(\frac{\zeta_{i+1}^{(j)}}{\zeta_i^{(j)}}\right)^2, \tag{54}$$

$$u_{i+1}^{(j)} = u_i^{(j)} + a_i^{(j)} d_i^{(j)}, \tag{55}$$

$$d_{i+1}^{(j)} = \zeta_{i+1}^{(j)} r_{i+1} + \beta_i^{(j)} d_i^{(j)}. \tag{56}$$

Here and below, Krylov identities are stated in exact arithmetic. A pole freezes once $|\zeta_{i+1}^{(j)}|\|r_{i+1}\| \le \delta_j$; subsequent iterations omit its vector and norm updates.

Proposition 3 (Pole sharing and three-scalar norm tracking)**.** *All shifted systems share the seed SpMVs. Their solution and search vectors need only be stored on V_C. The full solution norms are recovered by three scalars per pole, $h_i^{(j)} = \|u_i^{(j)}\|^2$, $c_i^{(j)} = u_i^{(j)\top} d_i^{(j)}$, and $v_i^{(j)} = \|d_i^{(j)}\|^2$, initialized by $h_0^{(j)} = c_0^{(j)} = 0$ and $v_0^{(j)} = \|w\|^2$. Suppressing j, their updates are*

$$h_{i+1} = h_i + 2a_i^{(j)} c_i + (a_i^{(j)})^2 v_i, \tag{57}$$

$$c_{i+1} = \beta_i^{(j)}(c_i + a_i^{(j)} v_i), \tag{58}$$

$$v_{i+1} = (\zeta_{i+1}^{(j)})^2\|r_{i+1}\|^2 + (\beta_i^{(j)})^2 v_i. \tag{59}$$

Proof. Binomial expansion gives $\mathcal{K}_i(A + \theta_j I, w) \subseteq \mathcal{K}_i(A, w)$; applying it to $A = (A + \theta_j I) - \theta_j I$ gives the reverse inclusion. Let P_i be the seed residual polynomial, so $r_i = P_i(A)w$ and $P_i(0) = 1$. Its roots are positive Ritz values. Therefore $P_i(-\theta_j) > 0$, and $Q_i(z) = P_i(z-\theta_j)/P_i(-\theta_j)$ has $Q_i(0) = 1$. The residual $Q_i(A+\theta_j I)w$ is a multiple of r_i, hence orthogonal to the shared $\mathcal{K}_i$. Moreover, $(1 - Q_i(z))/z$ is a polynomial of degree at most $i - 1$, so the corresponding iterate belongs to that space. Uniqueness of the SPD Galerkin iterate proves $r_i^{(j)} = \zeta_i^{(j)} r_i$ with $\zeta_i^{(j)} = 1/P_i(-\theta_j)$.

Eliminating d_i from seed CG gives $P_{i+1}(z) = (1 + \chi_i - a_i z)P_i(z) - \chi_i P_{i-1}(z)$. Evaluation at $-\theta_j$ gives the stated ζ recurrence. Matching the coefficient of z in the shifted recurrence gives $a_i^{(j)}$; taking the ratio of successive residual norms gives $\beta_i^{(j)}$. Thus (56) produces the shifted CG iterates without another SpMV. Since $P_i(-\theta_j) = \prod_\ell (1 + \theta_j/\lambda_\ell^{(i)}) \ge 1$, all shifted residuals meet a common tolerance when the seed does.

CG orthogonality gives $r_{i+1}^{(j)} \perp \mathcal{K}_{i+1}$, which contains both $u_{i+1}^{(j)}$ and $d_i^{(j)}$. Expanding the squared norm of $u_i^{(j)} + a_i^{(j)} d_i^{(j)}$ proves the first tracker identity. Taking its inner product with $r_{i+1}^{(j)} + \beta_i^{(j)} d_i^{(j)}$ proves the second. Expanding the squared norm of this search direction and using $r_{i+1}^{(j)\top} d_i^{(j)} = 0$ proves the third. These scalar recurrences use the full seed residual norm, while every shifted vector recurrence may be restricted to V_C coordinatewise. □

With at most t iterations and m current graph edges, one right-hand side therefore costs t SpMVs and $O(t(m+n+pn_C))$ arithmetic before freezing savings. The iteration count depends on the seed condition number, at most $(\Lambda + \tau_1)/(\sigma + \tau_1)$. Norm-tracking and recursive-residual roundoff contribute to the numerical score-error budget in Theorem 3.

B. Cross-Round Endpoint Maintenance

Let $W_S = C_S^\top U$ for the universal incidence sketch from Corollary 1, and maintain $X_j = (M_S + \tau_j I)^{-1} Z$ and $Y_j = (M_S + \tau_j I)^{-1} W_S$ only on V_C. The coarse score uses $h_{ej} = \|X_j^\top b_e\|^2$ and $t_{ej} = \|Y_j^\top b_e\|^2$. When edge $b = b_{e_r}$ is added, let $g_r \in \mathbb{R}^q$ be its row in U, so $W_{S+e_r} = W_S + b g_r^\top$. Put $u_j = (M_S + \tau_j I)^{-1} b$, $D_j = 1 + b^\top u_j$, $\xi_j = X_j^\top b$, and $\eta_j = g_r - Y_j^\top b$.

Theorem 4 (Slice maintenance and residual accumulation)**.** *The exact updates are*

$$X_j' = X_j - u_j \xi_j^\top / D_j, \qquad Y_j' = Y_j + u_j \eta_j^\top / D_j. \tag{60}$$

They can be evaluated solely on candidate endpoint rows in $O(pn_C q)$ time. For one pole, set $A_r = M_r + \tau I$, $a = \sigma + \tau$, and use approximate $\widetilde{u}_r$ with $f_r = A_{r-1}\widetilde{u}_r - b_r$ and $\|f_r\|_2 \le \delta$. Define $F_r = A_r \widetilde{X}_r - Z$ and update using $\widetilde{D}_r = 1 + b_r^\top \widetilde{u}_r$. If $k\delta \le \sqrt{a}$, then

$$\|\widetilde{X}_k - X_k\|_F \le \frac{e}{a}\|F_0\|_F + \frac{ek\delta}{a^{3/2}}\|Z\|_F. \tag{61}$$

In particular, exact initialization gives the second term alone.

Proof. Sherman–Morrison gives $(A + bb^\top)^{-1} = A^{-1} - uu^\top/(1 + b^\top u)$. Multiplication by Z proves the first identity. Multiplication by $W_S + bg_r^\top$, using $u^\top W_S = b^\top Y$ and $1 - (b^\top u)/(1 + b^\top u) = 1/(1 + b^\top u)$, proves the second. Restricting these outer products to V_C is exact because every queried b_e is supported there.

For approximate solves, $A_r\widetilde{u}_r = b_r\widetilde{D}_r + f_r$. Multiplying the computed X update by A_r therefore gives the exact residual recursion

$$F_r = F_{r-1} - f_r \frac{b_r^\top \widetilde{X}_{r-1}}{\widetilde{D}_r}. \tag{62}$$

Let $d = b_r^\top A_{r-1}^{-1} b_r > 0$ and $h = \delta/\sqrt{a} \le 1$. Since $A_{r-1} \succeq aI$,

$$|b_r^\top A_{r-1}^{-1} f_r| \le h\sqrt{d}, \qquad \|b_r^\top A_{r-1}^{-1}\|_2 \le \sqrt{d/a}.$$

Thus $\widetilde{D}_r \geq 1 + d - h\sqrt{d} \geq 3/4 > 0$, and the inequality $1 + d \geq 2\sqrt{d}$ yields

$$\frac{\|b_r^\top A_{r-1}^{-1}\|_2}{\widetilde{D}_r} \leq \frac{1}{\sqrt{a}(2-h)} \leq \frac{1}{\sqrt{a}}.$$

Because $\widetilde{X}_{r-1} = A_{r-1}^{-1}(Z + F_{r-1})$, (62) implies

$$\|F_r\|_F \leq (1+h)\|F_{r-1}\|_F + h\|Z\|_F.$$

Induction gives $\|F_k\|_F \leq (1+h)^k\|F_0\|_F + [(1+h)^k - 1]\|Z\|_F$. Finally, $\widetilde{X}_k - X_k = A_k^{-1}F_k$, $\|A_k^{-1}\| \leq 1/a$, and $kh \leq 1$ implies $(1+h)^k \leq e$ and $(1+h)^k - 1 \leq ekh$. These inequalities prove (61). □

If a rank-one update incurs an additional matrix perturbation E_r, its residual recurrence gains $A_r E_r$; the same induction adds $(e/a)\sum_{r=1}^{k}\|A_r E_r\|_F$ to (61).

The changing right-hand side of Y is handled explicitly. With $F_r^Y = A_r\widetilde{Y}_r - W_r$ and $\widetilde{\eta}_r = g_r - \widetilde{Y}_{r-1}^\top b_r$, multiplication of its update gives

$$F_r^Y = F_{r-1}^Y + f_r\widetilde{\eta}_r^\top/\widetilde{D}_r.$$

The inequalities in the preceding proof yield

$$\|F_r^Y\|_F \leq (1+h)\|F_{r-1}^Y\|_F + h\|W_{r-1}\|_F + \tfrac{4}{3}\delta\|g_r\|_2.$$

Writing $W_{\max} = \max_{r<k}\|W_r\|_F$ and summing gives

$$\|\widetilde{Y}_k - Y_k\|_F \leq \frac{e}{a}\left(\|F_0^Y\|_F + khW_{\max} + \frac{4\delta}{3}\sum_{r=1}^{k}\|g_r\|_2\right). \tag{63}$$

C. Shared-Direction Algebraic Norm Caching

Using the old slices and the vectors ξ_j, η_j from (60), we update all candidate norms by sharing two matrix–vector products.

Proposition 4 (Shared-direction norm cache)**.** *For every remaining candidate e, let*

$$h_{ej} = \|X_j^\top b_e\|^2, \qquad t_{ej} = \|Y_j^\top b_e\|^2, \qquad c_{ej} = b_e^\top u_j/D_j.$$

Compute the two shared directions $w_j^X = X_j\xi_j$ and $w_j^Y = Y_j\eta_j$. Then the exact cache updates are

$$h_{ej}^+ = h_{ej} - 2c_{ej}b_e^\top w_j^X + c_{ej}^2\|\xi_j\|^2, \tag{64}$$

$$t_{ej}^+ = t_{ej} + 2c_{ej}b_e^\top w_j^Y + c_{ej}^2\|\eta_j\|^2. \tag{65}$$

Maintaining both sketch slices and all candidate caches costs $O(pn_{\mathcal{C}}q + pN)$ per selected edge.

Proof. By (60), with $x = X_j^\top b_e$ and $y = Y_j^\top b_e$,

$$x^+ = x - c_{ej}\xi_j, \qquad y^+ = y + c_{ej}\eta_j.$$

Expanding these two squared norms gives (64)–(65), since $\langle x, \xi_j\rangle = b_e^\top X_j\xi_j$ and $\langle y, \eta_j\rangle = b_e^\top Y_j\eta_j$. Only rows in $V_{\mathcal{C}}$ are needed for w_j^X, w_j^Y, and each remaining quadratic-form coefficient is an endpoint difference. Computing the shared directions and applying the slice updates therefore costs $O(pn_{\mathcal{C}}q)$; all candidate scalar recurrences cost $O(pN)$. □

The cached oracle reads $O(p)$ scalars per candidate after initialization. The lazy oracle materializes q-dimensional norms only for competitive candidates. Our GPU implementation uses the cache backend.

D. Lossless Non-Submodular Lazy Selection

Put $a_j = \sigma + \tau_j$. The fixed-sketch marginal is

$$\widehat{G}(e \mid S) = \sum_j \omega_j \frac{h_{ej}}{1 + t_{ej} + a_j h_{ej}}. \tag{66}$$

We analyze lossless selection for this fixed proxy in exact arithmetic; residual and roundoff intervals provide enclosing keys in finite precision. All comparisons use the same lexicographic tie rule, with the maximum of an empty heap defined as $-\infty$.

At update ℓ, let

$$\chi_{\ell j} = \frac{\max_{v\in V_{\mathcal{C}}} u_{\ell j}(v) - \min_{v\in V_{\mathcal{C}}} u_{\ell j}(v)}{D_{\ell j}}.$$

Thus $|c_{e\ell j}| \leq \chi_{\ell j}$ for every candidate. Define global prefixes

$$A_j(r) = \sum_{\ell=1}^{r}\chi_{\ell j}\|\xi_{\ell j}\|, \qquad B_j(r) = \sum_{\ell=1}^{r}\chi_{\ell j}\|\eta_{\ell j}\|.$$

If edge e was last evaluated at round r_0, store $\nu_{ej} = \sqrt{h_{ej}(r_0)}$ and $\ell_{ej} = \sqrt{t_{ej}(r_0)}$, together with its prefix anchors. At round r, set

$$H_{ej}(r) = \nu_{ej} + A_j(r) - A_j(r_0), \tag{67}$$

$$L_{ej}(r) = \left[\ell_{ej} - B_j(r) + B_j(r_0)\right]_+, \tag{68}$$

$$U_e(r) = \sum_j \omega_j \frac{H_{ej}(r)^2}{1 + L_{ej}(r)^2 + a_j H_{ej}(r)^2}. \tag{69}$$

A tighter backend replaces each global increment $\chi_{\ell j}$ by $|c_{e\ell j}|$, maintaining the corresponding per-edge sums in $O(pN)$ time per round.

Theorem 5 (Envelope and common-affine heap)**.** *The bound $U_e(r) \geq \widehat{G}(e \mid S_r)$ holds for every remaining candidate. It costs $O(p)$ to query. Define*

$$D_\ell^{\mathrm{heap}} = \sum_j \omega_j\chi_{\ell j}\left(\frac{\|\xi_{\ell j}\|}{\sqrt{a_j}} + \frac{\|\eta_{\ell j}\|}{4a_j}\right), \qquad P(r) = \sum_{\ell=1}^{r} D_\ell^{\mathrm{heap}}. \tag{70}$$

A candidate evaluated at r_0 has the persistent key $K_e = \widehat{G}(e \mid S_{r_0}) - P(r_0)$, whose current upper bound is $K_e + P(r)$. A max-heap ordered by K_e, with the refinements in Algorithm 1, returns exactly the same proxy top-B set as full reevaluation, under the same tie-breaking rule.

Proof. Iterating (60) gives

$$x_e(r) = x_e(r_0) - \sum_{\ell=r_0+1}^{r} c_{e\ell j}\xi_{\ell j},$$

$$y_e(r) = y_e(r_0) + \sum_{\ell=r_0+1}^{r} c_{e\ell j}\eta_{\ell j}.$$

The triangle and reverse triangle inequalities imply $\|x_e(r)\| \le H_{ej}(r)$ and $\|y_e(r)\| \ge L_{ej}(r)$. For $h, t \ge 0$, the function $\phi(h,t) = h/(1+t+a_j h)$ has derivatives $\partial_h \phi = (1+t)/(1+t+a_j h)^2 > 0$ and $\partial_t \phi = -h/(1+t+a_j h)^2 \le 0$. Substituting the norm bounds and summing with positive weights proves (69). The global prefixes leave $O(p)$ arithmetic per queried edge.

For the persistent heap, regard $\phi(x,y) = \|x\|^2/(1+\|y\|^2 + a_j\|x\|^2)$. Let $z = \sqrt{a_j}\|x\|/\sqrt{1+\|y\|^2}$. Then

$$\|\nabla_x \phi\| = \frac{2z}{\sqrt{a_j}\sqrt{1+\|y\|^2}(1+z^2)^2} \le \frac{1}{\sqrt{a_j}},$$

where $2z \le 1+z^2$ was used. Writing $v = a_j\|x\|^2$ and $b = 1+\|y\|^2$, the inequality $v/(b+v)^2 \le 1/(4b)$ gives

$$\|\nabla_y \phi\| \le \frac{\|y\|}{2a_j(1+\|y\|^2)} \le \frac{1}{4a_j}.$$

Integrating these gradient bounds along the line segment joining the old and updated pair (x,y) yields

$$|\widehat{G}(e \mid S_\ell) - \widehat{G}(e \mid S_{\ell-1})| \le D_\ell^{\mathrm{heap}}.$$

Summing over updates proves $\widehat{G}(e \mid S_r) \le K_e + P(r)$. The additive term is common to every edge, so heap order is unchanged when P increases.

An envelope refinement at round r may replace K_e by $\min\{K_e, U_e(r) - P(r)\}$; the key remains an upper bound both now and after future common-prefix increments. An exact proxy evaluation replaces it by $\widehat{G}(e \mid S_r) - P(r)$. When such a current-round entry is maximal in the heap, its actual proxy score is at least every remaining upper bound, and therefore at least every remaining score. Extracting B current-round entries proves the top-B assertion. Reinserting the unselected shortlist members before increasing P preserves the same invariant for the next round. □

E. Zero-Waste Algebraic Re-ranking and Cost

For each shortlisted edge, a single-right-hand-side multi-shift solve supplies $u_j = R_j b_e$ and its tracked squared norm. The score

$$G_p(e \mid S) = \sum_j \omega_j \frac{\|u_j\|^2}{1 + b_e^\top u_j}$$

contains no projection variance. The champion's already computed slice is reused in (60). Thus initialization requires $2q$ multi-shift solves and each round requires at most B, giving at most $(2q + kB)t$ SpMVs. The shortlist contains the proxy maximizer, which is precisely the condition required by Theorem 3. The lazy backend and cache backend share this selection rule; the latter refines a heap entry by reading its scalar caches instead of computing q-dimensional candidate norms.

a) Total cost: Let H count heap operations and E count proxy refinements over all rounds. With cached norms, the arithmetic cost is

$$O\big(T_{\mathrm{proj}} + (2q+kB)t(m_0 + k + n + pn_{\mathcal{C}}) + pNq + k(pn_{\mathcal{C}}q + pN) + Ep + H(p + \log N)\big). \tag{71}$$

Algorithm 1: POLAR with cached or lazily materialized proxy norms

Input: $M_0, \mathcal{C}, k, s, \sigma, q, B$; backend cache or lazy
Output: Selected set S_k

/* Initialization */
Construct positive poles (ω_j, τ_j) and fixed normalized sketches Z, U
Extract $V_{\mathcal{C}}$ and candidate sketch rows g_e; set $S = \varnothing$, $M = M_0$
Use $2q$ multi-shift solves to initialize slices $X_j = R_j Z$ and $Y_j = R_j C_0^\top U$
Evaluate initial candidate norms and proxies; initialize anchors, heap keys $K_e = \widehat{G}(e \mid \varnothing)$, and stamps $r_e = 0$
Set all global prefixes A_j, B_j, P to zero
for $r = 0, \dots, k-1$ **do**
  /* Lossless proxy shortlist */
  $B_r \leftarrow \min\{B, N-r\}$; $\mathcal{B} \leftarrow \varnothing$
  while $|\mathcal{B}| < B_r$ **do**
    $(K_e, e) \leftarrow \mathrm{PopMax}(\mathcal{H})$
    if $r_e = r$ **then**
      $\mathcal{B} \leftarrow \mathcal{B} \cup \{e\}$
    else
      Compute U_e from (69)
      if $U_e - P < \mathrm{MaxKey}(\mathcal{H})$ **then**
        Push $(\min\{K_e, U_e - P\}, e)$
      else
        Obtain current h_{ej}, t_{ej} from scalar caches (cache) or slice norms (lazy)
        Refresh anchors and $r_e \leftarrow r$; push $(\widehat{G}(e \mid S) - P, e)$
  /* Algebraic re-ranking with tracked full norms */
  foreach $e \in \mathcal{B}$ **do**
    Run one multi-shift CG solve with early pole freezing and the three-scalar norm tracker; obtain u_{ej} and $\|u_{ej}\|^2$
    Compute the algebraic score, with the residual certificate of Lemma 5
    Keep the best-so-far score and slices; discard each losing contender
  Let a be the final champion, with computed slices u_{aj}
  Reinsert $\mathcal{B} \setminus \{a\}$ using their current-round keys
  /* Champion reuse and shared-direction caching */
  if $r + 1 < k$ **then**
    Compute $\xi_j, \eta_j, D_j, \chi_{r+1,j}$ from the old slices
    if *backend is* cache **then**
      Form w_j^X, w_j^Y and update all candidate scalar norms by (64)–(65)
    Apply the rank-one slice updates (60)
    Advance A_j, B_j and the common prefix P using (70)
  $S \leftarrow S \cup \{a\}$; update the sparse graph matrix $M \leftarrow M + b_a b_a^\top$
return S

The lazy oracle replaces Ep by Epq and can use global prefix envelopes without updating all candidate caches. Here $T_{\mathrm{proj}} = O((n + m_0 + N)q)$ for independent Gaussian entries and $O((n + m_0 + N)q^2)$ for the Gaussian–QR orthogonalized constructions. Storage for the direct two-bank implementation is $O((n + m_0 + N)q + pn_{\mathcal{C}}q + pN)$. Unpreconditioned CG has $t = O(\sqrt{\kappa_*}\log(\kappa_*/\mathrm{tol}))$ for a relative residual target. Pole freezing reduces vector updates while preserving the shared seed iteration count.

V. Experimental Evaluation

We address four questions. How close is POLAR to exhaustive search and to exact fractional greedy? How does its

running time scale on networks with millions of vertices? Does the exponent s change the links that are recommended and the diffusion behavior they produce? How sensitive is POLAR to its parameters?

A. Setup

a) Datasets: We use eighteen undirected real networks from KONECT [22] and SNAP [23]. Four small networks support exhaustive search: Zachary ($n = 34$, $m = 78$), Dolphins (62, 159), LesMis (77, 254), and Polbooks (105, 441). Five medium networks support exact greedy: Hamster (921, 4032), email-Eu-core (1005, 16 064), ca-GrQc (5242, 14 484), ca-HepPh (12 008, 118 489), and ca-CondMat (23 133, 93 439). Nine large networks appear in Table I; the largest is com-LiveJournal with 4.0 million vertices and 34.7 million edges. Directions, duplicate edges, and self-loops are removed, and all vertices are retained; graphs need not be connected.

b) Candidates and parameters: Unless stated otherwise, $\sigma = 1$, $k = 20$, and the candidate set contains $N = 2000$ missing links drawn uniformly at random with a seed fixed per graph. A *2-hop* pool instead samples missing friend-of-friend pairs by random two-step walks, the usual source of link recommendations. Small networks use $N = 40$ uniform candidates. POLAR uses $q = 128$ block-orthogonal sketches, shortlist $B = 16$, quadrature tolerance $\epsilon = 10^{-2}$ (between $p = 72$ and $p = 152$ poles, growing with $\log \kappa_*$), and relative CG tolerance 10^{-7}; the projection seed is 2026.

c) Baselines: Exact greedy maximizes the exact marginal of the target objective: candidates are screened with certified intervals from Theorem 1 at $\epsilon = 10^{-9}$, and every surviving contender is evaluated by a dense eigensolver. Proposition 5 below resolves the ties among structurally equivalent candidates that are frequent in 2-hop pools. *Exact-ER* and *Exact-Logdet* are exact greedy for $\mathrm{Tr}(M^{-1})$ and $\log\det M$; Exact-ER is the greedy augmentation of [2] with exact marginals. *Sketched-ER* and *Sketched-Logdet* run the POLAR machinery with the corresponding single-pole marginal. *SLQ* greedy estimates every candidate's trace change by stochastic Lanczos quadrature [24] with 20 Rademacher probes and 40 Lanczos steps. *Naive* recomputes both sketch banks in every round and selects the proxy maximizer. *Degree* repeatedly adds the candidate with the smallest current endpoint-degree sum, and *Random* averages ten seeded draws. Final gains are computed from eigenvalues when $n \le 6000$ and otherwise by certified quadrature with $\epsilon = 10^{-9}$ and CG tolerance 10^{-10}.

Proposition 5 (Twin candidates)**.** *Let u, v be nonadjacent in the current graph G_S and have the same weighted neighborhood. Then $g_s(\{u, v\} \mid S) = x^{-s} - (x + 2)^{-s}$ with $x = (M_S)_{uu}$.*

Proof. Equal neighborhoods give $(L_0 + L_S)b = d_u b$ for $b = \boldsymbol{e}_u - \boldsymbol{e}_v$, where d_u is the weighted degree in $G_S = G_0 + S$, so b is an eigenvector of M_S with eigenvalue $x = d_u + \sigma$. Adding $bb^\top$ moves this eigenvalue to $x + 2$ and leaves the orthogonal complement unchanged. □

d) Implementation: POLAR, Naive, SLQ, and the sketched baselines run on one NVIDIA RTX 4090 with CuPy 14.2 and a warp-per-row CSR kernel for multi-vector SpMM. It uses the cache backend and obtains the proxy top-B set by a dense sort on the GPU, which returns the same shortlist as the heap of Theorem 5. Exact greedy uses the GPU eigensolver for $n \le 6000$ and multithreaded CPU LAPACK for ca-HepPh and ca-CondMat. Timers include initial sketch solves, maintenance, scoring, and re-ranking, and exclude graph loading and sketch generation. Table I reports medians of three runs with rotated method order (two on web-Google and as-Skitter, one on LiveJournal). Naive runs three times on the five smallest networks and once elsewhere, with a one-hour cap checked between rounds. Code, data, and all raw results are available online [25].

B. Efficiency

Table I shows that endpoint maintenance removes most linear-algebra work. Naive solves $2kq = 5120$ right-hand sides, whereas POLAR solves $2q + k = 276$ with $B = 1$ and $2q + kB = 576$ with $B = 16$. Whenever Naive completes, SpMVs fall by 18.7–19.2× for $B = 1$, close to the ratio $5120/276 = 18.6$ of right-hand sides. On the eight networks where Naive finishes, $B = 1$ is 8.9–17.8× faster and returns exactly Naive's selections. The elapsed-time speedup approaches the SpMV ratio as graphs grow, because sparse products then dominate candidate scoring. The shortlist with $B = 16$ costs 1.2–2.5× the time of $B = 1$ and buys the quality reported below. Naive needs 194, 510, and 838 s per round on com-Youtube, as-Skitter, and com-LiveJournal, so a complete run would take 1.1, 2.8, and 4.7 hours; POLAR with $B = 1$ finishes in 3.6, 9.5, and 15.1 minutes, i.e., 18.1, 17.9, and 18.5× faster than the projected Naive runs. The $2q$ initial solves account for most of the remaining work: on web-Google, one Naive round, which equals one initialization, uses 94% of the SpMVs of the entire $B = 1$ run.

C. Effectiveness

a) Comparison with the optimum: On the four small networks, we enumerate all $\binom{40}{k}$ sets for $k = 1, \ldots, 5$ and $s \in \{0.25, 0.5\}$; the largest instance has 658 008 sets. In all forty instances, POLAR and exact fractional greedy attain the exhaustive optimum up to a relative difference of 2×10^{-11}. Exact-ER attains at least 0.9431 of it, Degree at least 0.8026, and Random 0.44–0.73 on average. The non-submodularity of Example 1 therefore does not prevent greedy selection from being optimal at these scales.

b) Comparison with exact greedy: Table II normalizes each method by exact greedy for the same exponent. POLAR attains 0.9920–1.0000 on uniform pools and 0.9974–1.0000 on 2-hop pools. SLQ greedy attains only 0.9327–0.9949: it forms each marginal as a difference of two stochastic trace estimates with absolute errors, whereas POLAR approximates the marginal itself with relative accuracy. Measured in the same runs, SLQ takes 12–39× the time of POLAR. Exact greedy takes up to 134 s on ca-GrQc and 1059–1096 s on ca-HepPh with CPU LAPACK, against 2.2 and 5.6 s for POLAR

TABLE I

RUNNING TIME (SECONDS) AND SPMV COUNTS ON THE ELEVEN LARGEST NETWORKS: $s = 0.5$, $k = 20$, $N = 2000$, $q = 128$, $p = 72$–152. SPEEDUPS ARE RELATIVE TO NAIVE. A TRUNCATED NAIVE RUN REPORTS ITS ELAPSED TIME AND THE NUMBER OF COMPLETED ROUNDS; ITS SPEEDUP IS A LOWER BOUND. POLAR WITH $B = 1$ SELECTS THE SAME LINKS AS NAIVE IN EVERY ROUND THAT NAIVE COMPLETED.

			Time (s)					SpMVs ($\times 10^3$)		
Network	n	m	Naive	$B = 1$	Speedup	$B = 16$	Speedup	Naive	$B = 1$	$B = 16$
ca-CondMat	23 133	93 439	29.7	3.3	8.9×	3.8	7.8×	524	27.7	53.1
ca-HepPh	12 008	118 489	43.7	4.2	10.4×	5.6	7.8×	840	43.8	77.5
email-Enron	36 692	183 831	90.6	8.1	11.2×	10.7	8.5×	1236	65.0	121
com-Amazon	334 863	925 872	197.1	12.1	16.3×	20.1	9.8×	622	33.0	62.1
com-DBLP	317 080	1 049 866	193.7	12.2	15.9×	20.0	9.7×	647	34.5	66.6
roadNet-PA	1 088 092	1 541 898	139.6	8.0	17.4×	17.9	7.8×	131	7.0	13.6
roadNet-CA	1 965 206	2 766 607	268.7	15.1	17.8×	32.8	8.2×	136	7.2	13.8
com-Youtube	1 134 890	2 987 624	>3687 (19)	214.8	$> 17.2\times$	542.6	$> 6.8\times$	>3356	187	357
web-Google	875 713	4 322 051	2225.1	125.0	17.8×	303.0	7.3×	2776	147	278
as-Skitter	1 696 415	11 095 298	>4077 (8)	569.9	$> 7.2\times$	1389.5	$> 2.9\times$	>2619	348	656
com-LiveJournal	3 997 962	34 681 189	>2515 (3)	908.8	$> 2.8\times$	1720.9	$> 1.5\times$	>469	166	311

TABLE II

GAIN RELATIVE TO EXACT FRACTIONAL GREEDY FOR THE SAME EXPONENT: $k = 20$, $N = 2000$, $\sigma = 1$. EXACT-ER AND EXACT-LOGDET ARE EXACT GREEDY FOR OTHER OBJECTIVES, EVALUATED ON f_s. RANDOM IS THE MEAN OF TEN DRAWS. ON CA-GRQC, THE LEADING 2-HOP CANDIDATES ARE TIED TWIN PAIRS (PROPOSITION 5); UP TO 77 CANDIDATES SHARE THE MAXIMAL MARGINAL IN A ROUND, AND ALL NON-RANDOM METHODS ATTAIN THE SAME GAIN.

		Uniform pool							2-hop pool			
Network	s	POLAR	SLQ	Exact-ER	Exact-Logdet	Sketched-ER	Degree	Random	POLAR	Exact-ER	Exact-Logdet	Degree
email-Eu-core	0.25	0.9949	0.9630	0.9854	0.9942	0.9680	0.9207	0.245	0.9974	0.9669	0.9907	0.9174
	0.5	0.9954	0.9693	0.9967	0.9870	0.9854	0.8731	0.213	0.9989	0.9850	0.9708	0.8724
	0.75	0.9975	0.9949	1.0000	0.9753	0.9943	0.8213	0.189	0.9998	0.9973	0.9489	0.8264
Hamster	0.25	0.9964	0.9327	0.9965	1.0000	0.9965	0.8890	0.480	0.9998	0.9680	1.0000	0.9923
	0.5	1.0000	0.9541	0.9981	1.0000	0.9981	0.8497	0.434	0.9997	0.9819	1.0000	0.9899
	0.75	1.0000	0.9458	0.9997	0.9994	0.9997	0.8060	0.396	1.0000	0.9961	0.9977	0.9851
ca-GrQc	0.25	0.9920	0.9640	0.9944	1.0000	0.9914	0.8838	0.543	1.0000	1.0000	1.0000	1.0000
	0.5	0.9952	0.9541	0.9970	1.0000	0.9944	0.8456	0.500	1.0000	1.0000	1.0000	1.0000
	0.75	0.9981	0.9477	1.0000	0.9997	0.9981	0.8051	0.465	1.0000	1.0000	1.0000	1.0000
ca-HepPh	0.25	0.9942	–	0.9905	0.9973	0.9874	0.9300	0.441	1.0000	0.9754	1.0000	0.9963
	0.5	1.0000	–	0.9954	0.9871	0.9919	0.8927	0.401	1.0000	0.9852	1.0000	0.9952
	0.75	0.9997	–	0.9985	0.9714	0.9948	0.8476	0.367	1.0000	0.9963	0.9985	0.9927

in isolated runs (Tables V and I); on Hamster and email-Eu-core, dense exact greedy needs only 1–2 s and is the better choice. On ca-CondMat, where only $s = 0.5$ and Exact-ER were run on the CPU (6471 and 6575 s), the ratios are 0.9944 for POLAR and 0.9940 for Exact-ER.

With $\sigma = 1$, the alternative exact objectives remain close to the fractional optimum, but no single one is uniformly close. On 2-hop pools, Exact-ER loses up to 3.3%, Exact-Logdet up to 5.1%, and Degree up to 17.4% (19.5% on uniform pools). Their losses grow at the exponent farther from their own: Exact-ER is weakest at $s = 0.25$, and Exact-Logdet at $s = 0.75$. Conversely, exact ER or log-det greedy is slightly better than POLAR on f_s in 7 of the 12 uniform-pool cells of Table II, by at most 0.8% (ca-GrQc), and in 2 of the 12 2-hop cells, by at most 0.03%. In each such cell POLAR is also below exact fractional greedy, so the gap is sketch error, not an advantage of the other objective; at $s = 0.5$ on ca-GrQc, $B = 32$ closes it (Table V).

c) Large networks: Dense exact greedy is out of reach on the seven large networks of Table III (it already takes 1.8 h per objective on ca-CondMat), so each method is normalized by the best method for each target. The exponent-matched POLAR run attains 0.9979–1.0000 in sixteen of seventeen cases, while Degree attains 0.7677–0.9911. The exception is com-Youtube, where Sketched-ER is 1.4% better on $f_{0.5}$. Enlarging the shortlist to $B = 64$ raises POLAR to 1.0040 of the Sketched-ER gain (1072 s), and $q = 256$ raises it to 0.9948 (753 s). The gap is therefore a sketch ranking error at $q = 128$: the best candidate lies outside the proxy top-16, and the objective is not at fault.

D. Controlled Multiscale Benchmark

On the real networks above, objectives with $\sigma = 1$ attain similar f_s while selecting different links: at $\sigma = 1$, the Jaccard similarity between the exact $s = 0.25$ and Exact-ER selections is 0.43–0.74 on email-Eu-core, Hamster, and ca-GrQc. To isolate when the exponent matters, the graph in Fig. 1(b) contains two 390-vertex cores, each formed by three independently permuted Hamilton cycles with weight 10, a weight-0.5 intercore bridge, and twenty weight-0.3 leaves attached to the first core. Five further vertices form a weight-0.5 cycle, each with its own weight-0.5 tie to that core. Thus

TABLE III
GAIN RELATIVE TO THE BEST METHOD ON EACH TARGET FOR LARGE NETWORKS. "OTHER s" IS THE BEST POLAR RUN FOR A DIFFERENT EXPONENT, EVALUATED ON THE SAME TARGET.

Network	s	POLAR	Other s	Sk.-ER	Sk.-Logdet	Degree	Random
email-Enron	0.25	1.0000	0.9995	0.9935	0.9919	0.8706	0.570
	0.5	1.0000	0.9997	0.9980	0.9862	0.8229	0.525
	0.75	0.9988	0.9970	1.0000	0.9762	0.7677	0.485
com-DBLP	0.25	0.9990	1.0000	0.9819	0.9993	0.9911	0.555
	0.5	1.0000	0.9997	0.9910	0.9946	0.9824	0.517
	0.75	1.0000	0.9922	0.9959	0.9807	0.9641	0.483
com-Amazon	0.25	0.9979	0.9997	0.9821	1.0000	0.9798	0.512
	0.5	1.0000	0.9986	0.9886	0.9978	0.9692	0.470
	0.75	1.0000	0.9973	0.9946	0.9919	0.9536	0.438
roadNet-PA	0.25	0.9981	1.0000	0.9997	0.9987	0.9898	0.646
	0.5	0.9987	1.0000	0.9995	0.9980	0.9857	0.613
	0.75	1.0000	0.9983	0.9993	0.9970	0.9809	0.588
roadNet-CA	0.25	1.0000	1.0000	0.9756	1.0000	0.9885	0.650
	0.5	1.0000	1.0000	0.9833	1.0000	0.9840	0.617
	0.75	1.0000	0.9985	0.9934	0.9984	0.9773	0.591
web-Google	0.5	1.0000	–	0.9874	0.9914	0.9724	0.448
com-Youtube	0.5	0.9858	–	1.0000	0.9766	0.8728	0.658

$n = 805$, $\sigma = 10^{-2}$, and the shared pool contains 2995 sampled intercore or leaf-to-core links plus the five missing cycle chords; $k = 15$, and graph seeds 711–713 are fixed across methods.

Table IV connects gains to link types. Exact-ER assigns seven of fifteen links to the intercore cut. POLAR assigns all fifteen links to leaf repairs at $s = 0.25$, and fourteen leaf repairs and one intercore link at $s = 0.5$, matching exact fractional greedy. Each cycle vertex has weighted degree 1.5, so any unselected cycle chord has endpoint-degree sum at most 5, whereas leaf-to-core and intercore candidates have sums of at least 60.3 and 120. Degree therefore spends five links on the cycle, although the cycle already has five independent core attachments. At $s = 0.25$ and $s = 0.5$, POLAR improves the mean gain over Degree by 26.81% and 36.94%, over Exact-ER by 48.06% and 21.77%, and over Sketched-ER by 19.31% and 5.65%; it recovers 99.95% and 99.92% of exact fractional greedy. Without the five-vertex cycle ($n = 800$ control, seed 711), Degree and POLAR attain 6.077892 and 6.078605 at $s = 0.25$, so the cycle is what separates local degree from fractional spectral value.

E. Diffusion Time Scales

Proposition 1 predicts the order Degree, Exact-Logdet, $s = 0.25$, 0.5, 0.75, Exact-ER from short to long times. Since $s/\sigma \le 1$ for $\sigma = 1$, f_s weights the short times at which all selections behave alike, which is why their values of f_s are close even when their heat traces separate at $t \ge 1$.

Figure 2 confirms this order on ca-HepPh. At $t = 0.03$, Degree is best and Exact-ER attains 98.8% of it. At $t = 3.2$, Exact-ER is best, while Degree, Exact-Logdet, and POLAR with $s = 0.25$, 0.5, and 0.75 attain 38.2%, 76.3%, 86.2%, 94.8%, and 97.2%. On all five uniform pools, Degree is best at the shortest times, an inverse-trace selection at the longest, and the fractional selections lie between them in the order of s. On the ca-HepPh uniform pool, the heat-trace reductions at $t = 3.2$ differ by 24% between the log-determinant and inverse-trace

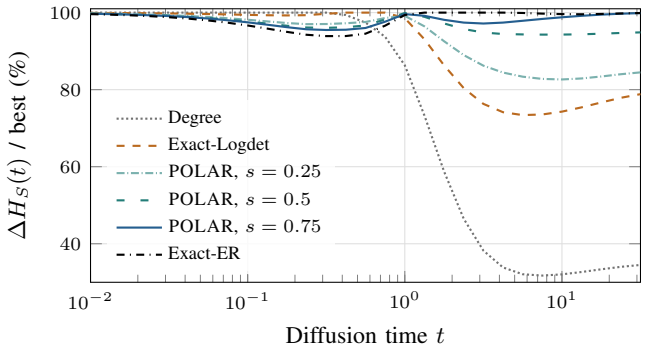


Fig. 2. Heat-trace reduction on ca-HepPh (uniform pool, $k = 20$), relative to the best of all selections at each time. POLAR at $s = 0.5$ selects the same set as exact greedy.

selections, although their f_s values differ by at most 3%, and by 62% between Degree and Exact-ER. The exponent thus orders recommendations by the diffusion horizon they improve. On 2-hop pools, few added links merge components, so $\Delta H_S(t) \to 0$ and ratios at $t \ge 30$ are dominated by rounding.

F. Parameters and Robustness

Table V varies one parameter at a time on ca-GrQc. The shortlist has the largest effect: $B = 1$ yields 0.9796, $B = 8$ already reaches 0.9952, and $B = 32$ matches the exact greedy gain, while time grows from 1.35 to 2.23 s. Recovery generally improves with q, from 0.9694 at $q = 16$ to 0.9999 at $q = 192$. The quadrature tolerance has no visible effect: the final gain is the same to four decimals while p grows from 60 to 110, so the relative marginal bound is conservative for ranking.

a) Sketch construction: Over five projection seeds with $q = 128$ and $B = 16$, block-orthogonal sketches attain mean ratios 0.9938–0.9997 on email-Eu-core, Hamster, and ca-GrQc, against 0.9887–0.9997 for Gaussian sketches. With $q = 24$ and $B = 4$, the block-orthogonal mean drops to 0.9480–0.9909. Real networks therefore need larger sketches than the multiscale benchmark, whose coarse ranking is already stable at $q = 24$.

b) Per-round recovery: Let r_i be the ratio of the chosen edge's true marginal to the largest true marginal in round i of the POLAR trajectory. With the defaults, the per-trajectory mean of r_i is 0.964–1.000 and its minimum is 0.900–1.000; the round winner is the true maximizer in 58–96% of rounds. The minimum enters β in Lemma 4, and the mean enters Remark 1. The worst-case factors are conservative (about 0.34 at $\sigma = 1$ and $s = 0.5$); the measured recovery shows that the practical gap is far smaller.

c) Candidate pools, budget, and regularization: Over three independently drawn uniform pools for each of email-Eu-core, Hamster, and ca-GrQc, POLAR attains 0.9889–1.0000 of exact greedy. On the ca-GrQc 2-hop pool with $k = 100$, POLAR with $s = 0.5$ attains 0.9929, against 0.9793 for Exact-ER, 0.9918 for Degree, and 0.9988 for Exact-Logdet. For $\sigma \in \{0.01, 0.1\}$, POLAR with $s = 0.5$ attains the exact greedy gain on all three networks. As σ decreases, the fractional selections approach Exact-ER, while Degree falls to 0.20–0.58 of the best $f_{0.5}$.

TABLE IV

TRUE FRACTIONAL GAINS ON THE MULTISCALE BENCHMARK: $n = 805$, $\sigma = 10^{-2}$, $N = 3000$, $k = 15$. GAINS ARE MEANS $\pm$ SAMPLE STANDARD DEVIATIONS OVER GRAPH SEEDS 711, 712, AND 713. POLAR USES GLOBAL COLUMN-ORTHOGONAL SKETCHES, $q = 24$, $B = 4$, AND $p = 20$. Δ_{ER} COMPARES MEAN GAINS WITH EXACT-ER. C/L/D DENOTES INTERCORE LINKS / LEAF REPAIRS / LOCAL CYCLE CHORDS. COUNTS ARE IDENTICAL ACROSS SEEDS EXCEPT RANDOM, WHOSE MEAN ALLOCATION IS SHOWN.

	$s = 0.25$			$s = 0.5$		
Method	True gain	Δ_{ER} (%)	C/L/D	True gain	Δ_{ER} (%)	C/L/D
Random	2.941226 ± 0.586256	-28.36	$10/5/0$	9.257625 ± 1.170745	-20.83	$10/5/0$
Degree	4.793361 ± 0.000284	$+16.75$	$0/10/5$	10.398292 ± 0.000542	-11.08	$0/10/5$
Exact-ER ($s = 1$)	4.105538 ± 0.000227	$+0.00$	$7/8/0$	11.694059 ± 0.000555	$+0.00$	$7/8/0$
Sketched-ER ($q = 24$)	5.094869 ± 0.000657	$+24.10$	$4/11/0$	13.477414 ± 0.001683	$+15.25$	$4/11/0$
POLAR	$\mathbf{6.078574 \pm 0.000183}$	$\mathbf{+48.06}$	$\mathbf{0/15/0}$	$\mathbf{14.239384 \pm 0.000217}$	$\mathbf{+21.77}$	$\mathbf{1/14/0}$
Exact fractional greedy	6.081501 ± 0.000393	$+48.13$	$0/15/0$	14.250901 ± 0.000144	$+21.86$	$1/14/0$

TABLE V

ONE-AT-A-TIME PARAMETER STUDY ON CA-GRQC, $s = 0.5$: GAIN RELATIVE TO EXACT GREEDY, TIME, AND POLE COUNT. DEFAULTS ARE $q = 128$, $B = 16$, $\epsilon = 10^{-2}$.

q	16	32	64	96	128	192	256
Ratio	.9694	.9801	.9761	.9934	.9952	.9999	.9987
Time (s)	1.64	1.66	1.83	1.95	2.17	2.48	2.64
B	1	2	4	8	16	32	64
Ratio	.9796	.9809	.9826	.9952	.9952	1.0000	1.0000
Time (s)	1.35	1.79	1.99	2.15	2.17	2.23	3.33
ϵ	10^{-1}	10^{-2}	10^{-3}	10^{-4}			
p	60	80	90	110			
Ratio	.9952	.9952	.9952	.9952			
Time (s)	2.02	2.17	2.24	2.44			

VI. RELATED WORK AND CONCLUSION

A. Related Work

a) Spectral design and graph mining: Effective-resistance minimization has been studied through convex edge-weight design [1] and greedy edge augmentation [2]. Current-flow group centrality [3], [4] and forest closeness [12] optimize vertex groups. Dynamic forest-matrix queries [5] and forest-based opinion optimization [13] further demonstrate the role of Laplacian resolvents in graph mining. Kemeny's constant has scalable estimators [26] and a distinct edge-addition theory [14]. The companion work FastFrac [11] maximizes the positive-power trace $\mathrm{Tr}((I+\alpha L)^s)$, which is submodular. Its quadrature reduction does not extend to negative exponents, where the objective is not submodular (Example 1) and marginals require squared resolvent norms; POLAR addresses this regime with sketched squared norms and exact non-submodular proxy shortlists.

b) Matrix functions and shifted Krylov methods: Contour quadrature [15] and rational Krylov methods for fractional graph diffusion [6] provide complementary approaches to matrix functions. Short-recurrence shifted solvers [9] and restarted shifted GMRES [27] share Krylov spaces across shifts. Lanczos stability and error bounds are developed in [18], [19], and stochastic Lanczos quadrature [24] estimates traces of matrix functions. Our analysis applies positive quadrature directly to rank-one trace marginals and combines it with cross-round endpoint maintenance.

c) Solvers and spectral approximation: Nearly linear SDD solvers [28], [29] and sparse approximate elimination [30] provide algorithmic foundations for Laplacian systems. Resistance sampling [20] and deterministic spectral sparsification [31] preserve graph quadratic forms with few edges.

d) Greedy selection and sketches: CELF exploits decreasing marginals [8]; the ratio–curvature framework handles monotone non-submodular objectives [10]. POLAR derives explicit fractional parameters and replaces stale marginal keys with algebraic envelopes. Orthogonal projections have sharp confidence bounds [21]; our trajectory-wide certificate covers adaptive selections with fixed sketches.

B. Conclusion

POLAR makes fractional inverse-trace link recommendation practical on graphs with tens of millions of edges. Positive quadrature gives a dimension-free marginal guarantee, ratio and curvature bounds support error-aware greedy selection, and pole sharing, endpoint maintenance, and algebraic re-ranking reduce k rounds to $2q + kB$ multi-shift solves. On eighteen real networks, POLAR matched exhaustive search on small instances, attained 99.2–100% of exact fractional greedy, and processed LiveJournal in 15 minutes with $B = 1$. The heat-trace representation explains what the exponent controls: smaller s favors links that reduce short-time return probabilities, larger s favors long-time mixing, and the selected links follow this order; with $\sigma = 1$ the objectives differ mainly in which links they pick, while multiscale structure also separates their values. Future work includes preconditioned solvers for high-degree networks and maintaining the sketches as the base graph evolves.

DECLARATION ON THE USE OF GENERATIVE AI TOOLS

Generative AI tools were used only for language polishing and grammar checking during the preparation of this manuscript. The authors reviewed and verified all AI-assisted edits and take full responsibility for the content of the manuscript.

REFERENCES

[1] A. Ghosh, S. Boyd, and A. Saberi, "Minimizing effective resistance of a graph," *SIAM Review*, vol. 50, no. 1, pp. 37–66, 2008, doi: 10.1137/050645452.

[2] M. Predari, L. Berner, R. Kooij, and H. Meyerhenke, "Greedy optimization of resistance-based graph robustness with global and local edge insertions," *Social Network Analysis and Mining*, vol. 13, Art. 130, 2023, doi: 10.1007/s13278-023-01137-1.

[3] H. Li, R. Peng, L. Shan, Y. Yi, and Z. Zhang, "Current flow group closeness centrality for complex networks," in *Proc. The Web Conference (WWW)*, 2019, pp. 961–971, doi: 10.1145/3308558.3313490.

[4] H. Xia and Z. Zhang, "Fast maximization of current flow group closeness centrality," in *Proc. IEEE ICDE*, 2025, doi: 10.1109/ICDE65448.2025.00241.

[5] H. Sun, X. Zhou, and Z. Zhang, "Fast computation for the forest matrix of an evolving graph," in *Proc. ACM SIGKDD*, 2024, pp. 2755–2764, doi: 10.1145/3637528.3671822.

[6] M. Benzi and I. Simunec, "Rational Krylov methods for fractional diffusion problems on graphs," *BIT Numerical Mathematics*, vol. 62, pp. 357–385, 2022, doi: 10.1007/s10543-021-00881-0.

[7] M. Black, Z. Wan, A. Nayyeri, and Y. Wang, "Understanding oversquashing in GNNs through the lens of effective resistance," in *Proc. ICML*, PMLR, vol. 202, 2023, pp. 2528–2547.

[8] J. Leskovec, A. Krause, C. Guestrin, C. Faloutsos, J. VanBriesen, and N. Glance, "Cost-effective outbreak detection in networks," in *Proc. ACM SIGKDD*, 2007, pp. 420–429.

[9] B. Jegerlehner, "Krylov space solvers for shifted linear systems," arXiv:hep-lat/9612014, 1996.

[10] A. A. Bian, J. M. Buhmann, A. Krause, and S. Tschiatschek, "Guarantees for greedy maximization of non-submodular functions with applications," in *Proc. ICML*, PMLR, vol. 70, pp. 498–507, 2017.

[11] "FastFrac: Fast edge recommendation for fractional spectral traces on graphs," companion submission to *IEEE ICDE*, 2027.

[12] A. van der Grinten, E. Angriman, M. Predari, and H. Meyerhenke, "New approximation algorithms for forest closeness centrality—for individual vertices and vertex groups," in *Proc. SIAM SDM*, 2021, pp. 136–144, doi: 10.1137/1.9781611976700.16.

[13] H. Sun, Y. Sun, X. Zhou, and Z. Zhang, "Fast computation and optimization for opinion-based quantities of Friedkin–Johnsen model," in *Advances in Neural Information Processing Systems*, vol. 38, 2025, doi: 10.52202/085713-1269.

[14] S. Kirkland, Y. Li, J. S. McAlister, and X. Zhang, "Edge addition and the change in Kemeny's constant," *Discrete Applied Mathematics*, vol. 373, pp. 77–90, 2025, doi: 10.1016/j.dam.2025.04.031.

[15] N. Hale, N. J. Higham, and L. N. Trefethen, "Computing A^α, $\log(A)$ and related matrix functions by contour integrals," *SIAM Journal on Numerical Analysis*, vol. 46, no. 5, pp. 2505–2523, 2008, doi: 10.1137/070700607.

[16] NIST, *Digital Library of Mathematical Functions*, Sec. 3.5, "Quadrature." [Online]. Available: https://dlmf.nist.gov/3.5

[17] NIST, *Digital Library of Mathematical Functions*, Sec. 18.3, "Definitions." [Online]. Available: https://dlmf.nist.gov/18.3

[18] C. Musco, C. Musco, and A. Sidford, "Stability of the Lanczos method for matrix function approximation," in *Proc. ACM–SIAM SODA*, 2018, pp. 1605–1624, doi: 10.1137/1.9781611975031.105.

[19] T. Chen, A. Greenbaum, C. Musco, and C. Musco, "Error bounds for Lanczos-based matrix function approximation," *SIAM Journal on Matrix Analysis and Applications*, vol. 43, no. 2, pp. 787–811, 2022, doi: 10.1137/21M1427784.

[20] D. A. Spielman and N. Srivastava, "Graph sparsification by effective resistances," *SIAM Journal on Computing*, vol. 40, no. 6, pp. 1913–1926, 2011, doi: 10.1137/080734029.

[21] M. Skorski, "Johnson–Lindenstrauss transforms with best confidence," in *Proc. COLT*, PMLR, vol. 134, pp. 3989–4007, 2021.

[22] J. Kunegis, "KONECT: The Koblenz network collection," in *Proc. WWW Companion*, 2013, pp. 1343–1350, doi: 10.1145/2487788.2488173.

[23] J. Leskovec and A. Krevl, "SNAP Datasets: Stanford large network dataset collection," June 2014. [Online]. Available: https://snap.stanford.edu/data/

[24] S. Ubaru, J. Chen, and Y. Saad, "Fast estimation of $\mathrm{tr}(f(A))$ via stochastic Lanczos quadrature," *SIAM Journal on Matrix Analysis and Applications*, vol. 38, no. 4, pp. 1075–1099, 2017, doi: 10.1137/16M1104974.

[25] POLAR: Code, data, results, and extended version with complete proofs. [Online]. Available: https://github.com/chenty2333/POLAR

[26] S. Li, X. Huang, and C.-H. Lee, "An efficient and scalable algorithm for estimating Kemeny's constant of a Markov chain on large graphs," in *Proc. ACM SIGKDD*, 2021, pp. 964–974, doi: 10.1145/3447548.3467431.

[27] A. Frommer and U. Glässner, "Restarted GMRES for shifted linear systems," *SIAM Journal on Scientific Computing*, vol. 19, no. 1, pp. 15–26, 1998, doi: 10.1137/S1064827596304563.

[28] D. A. Spielman and S.-H. Teng, "Nearly linear time algorithms for preconditioning and solving symmetric, diagonally dominant linear systems," *SIAM Journal on Matrix Analysis and Applications*, vol. 35, no. 3, pp. 835–885, 2014, doi: 10.1137/090771430.

[29] I. Koutis, G. L. Miller, and R. Peng, "A nearly-$m \log n$ time solver for SDD linear systems," in *Proc. IEEE FOCS*, 2011, pp. 590–598, doi: 10.1109/FOCS.2011.85.

[30] R. Kyng and S. Sachdeva, "Approximate Gaussian elimination for Laplacians—fast, sparse, and simple," in *Proc. IEEE FOCS*, 2016, pp. 573–582, doi: 10.1109/FOCS.2016.68.

[31] J. Batson, D. A. Spielman, and N. Srivastava, "Twice-Ramanujan sparsifiers," *SIAM Journal on Computing*, vol. 41, no. 6, pp. 1704–1721, 2012, doi: 10.1137/090772873.